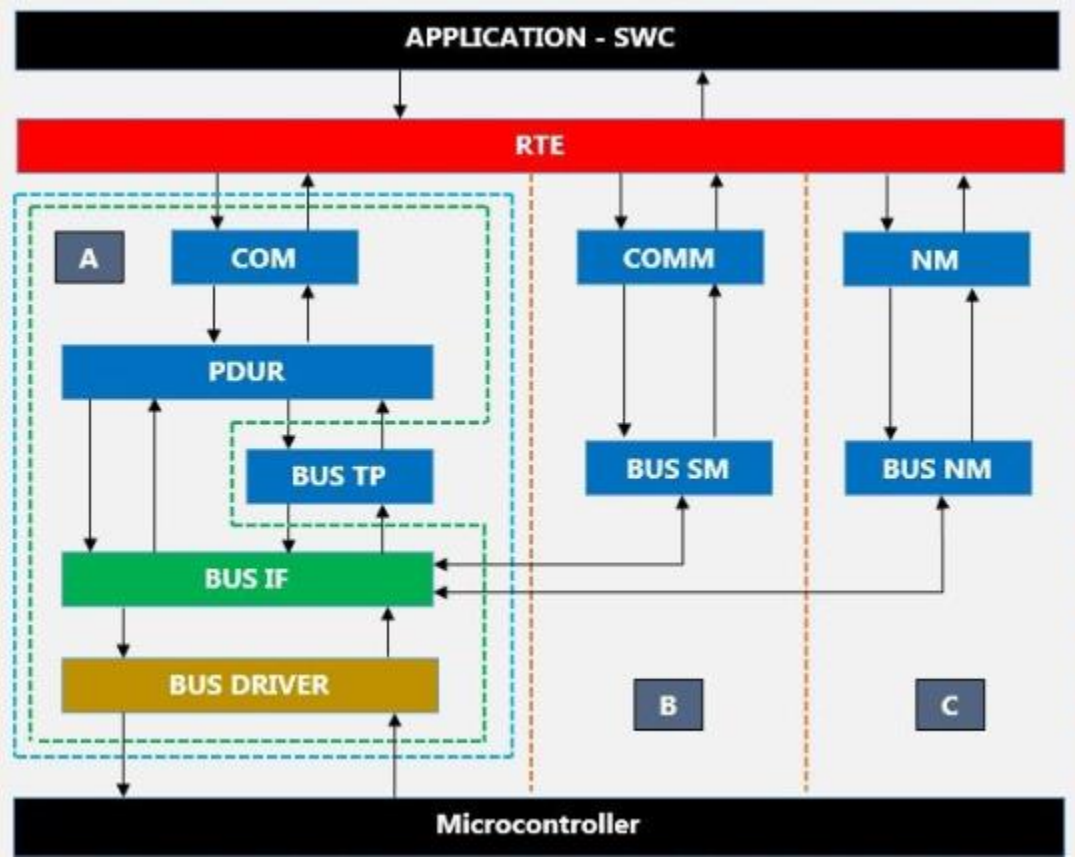
AUTOSAR COM Stack

Agenda

- Introduction
- CAN
- CAN Driver Module
- CAN IF Module
- CanTP Module
- PduR Module
- COM Module

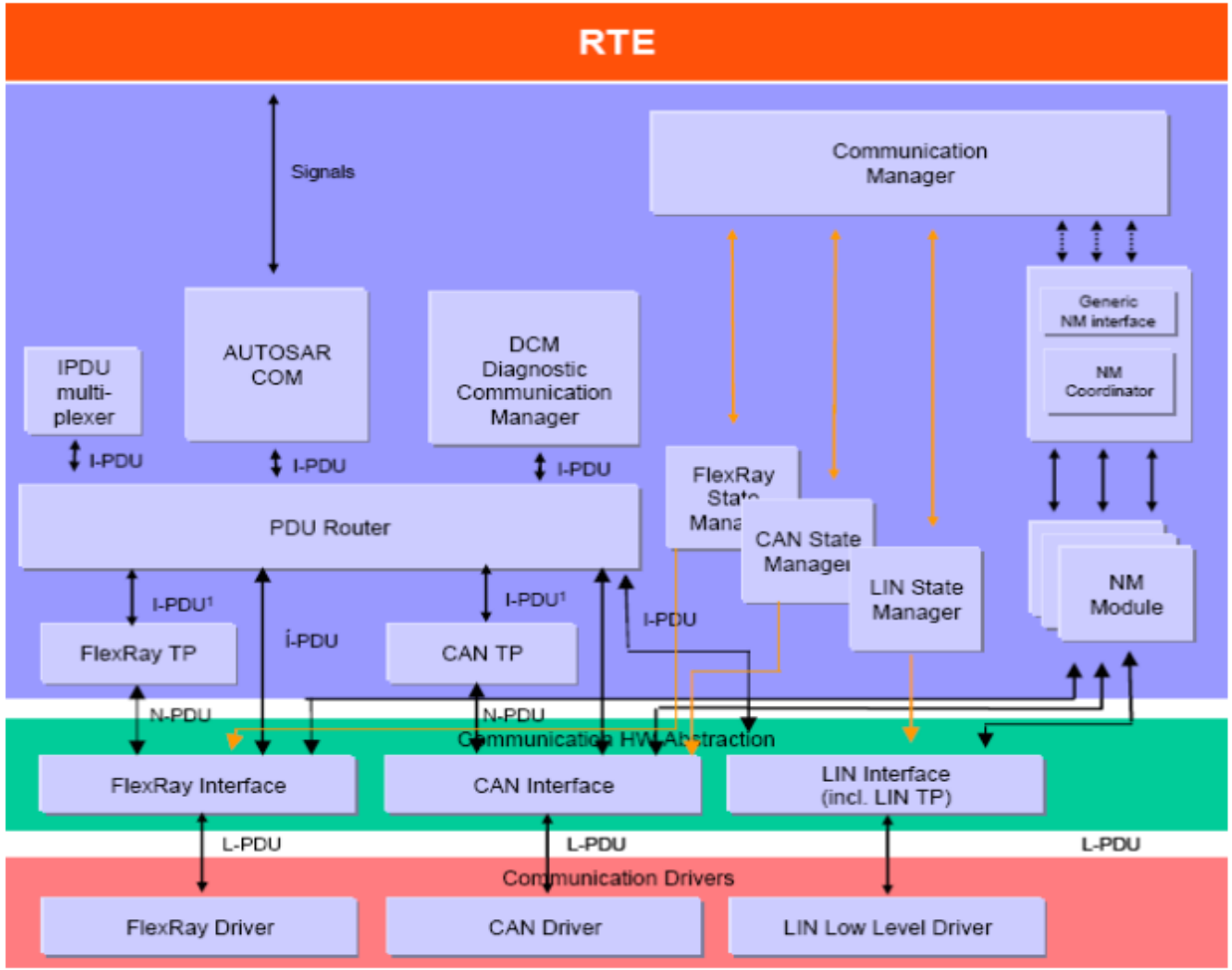
Com Stack



A. Data Transmission and reception Management

B. State Management

C. Network Mangagement


RTE
Signals
Communication Manager
AUTOSAR COM
DCM Diagnostic Communication Manager
IPDU multi-plexer
Generic NM interface
NM Coordinator
I-PDU
I-PDU
I-PDU
PDU Router
FlexRay Sta
Mana
CAN State Manager
LIN State Manager
NM Module
I-PDU¹
I-PDU¹
I-PDU
I-PDU
FlexRay TP
CAN TP
N-PDU
N-PDU
Communication HW Abstraction
FlexRay Interface
CAN Interface
LIN Interface (incl. LIN TP)
L-PDU
L-PDU
L-PDU
Communication Drivers
FlexRay Driver
CAN Driver
LIN Low Level Driver

Autosar COM Module

Main Features of COM

- Provision of signal oriented data interface for the users (e.g. RTE, SwC)
- Packing of AUTOSAR signals to I-PDUs to be transmitted
- Unpacking of received I-PDUs and provision of received signals to the users (e.g. RTE, SwC)
- Routing of signals from received I-PDUs into I-PDUs to become transmitted
- Routing of signal groups from received I-PDUs into I-PDUs to become transmitted
- Communication transmission control (start/ stop of I-PDU groups)
- Replications of send requests

- Guarantee of minimum distances between transmit I-PDUs
- Monitoring of receive signals (signals timeout)
- Filter mechanisms for incoming signals
- Different notification mechanisms
- Provision of init values and update indications
- Byte order conversion
- Sign extension
- Support of two different transmission modes per I-PDU
- Signal based gateway
- Support of large and dynamic length data types

Dependencies to Other Modules

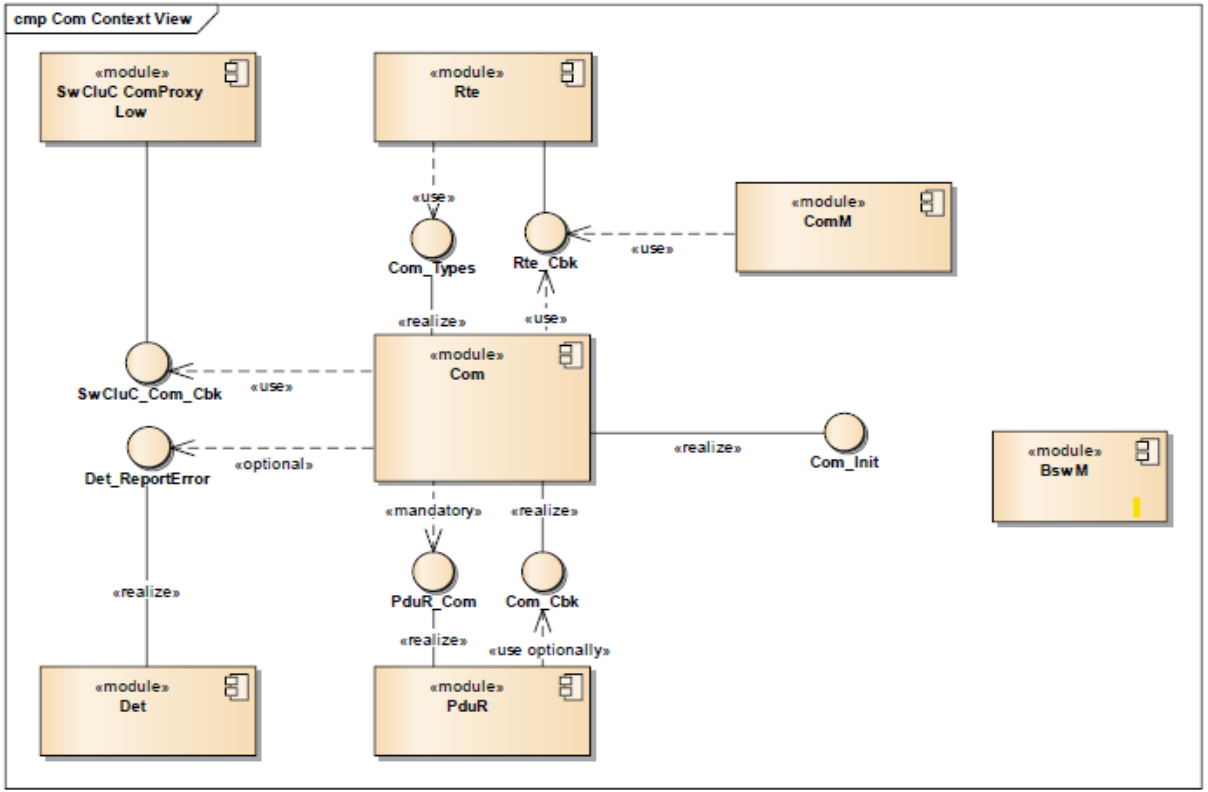


Figure 1: AUTOSAR COM module's context view

PDU Router

The AUTOSAR COM module uses the union of:

- APIs for upper layer modules that uses TP
- APIs for upper layer modules that do not use TP.

COM module transports I-PDUs either unfragmented via simple L-PDUs or fragmented via TP.

PDUR functionalities needed by COM:

- Indication of incoming I-PDUs
- Sending interface for outgoing I-PDUs including the confirmation if an I-PDU has been sent by the communication controller
- Trigger interface to enable the PDU router to cause a transmission from the AUTOSAR COM module
- Buffer handling for TP communication

Com Users

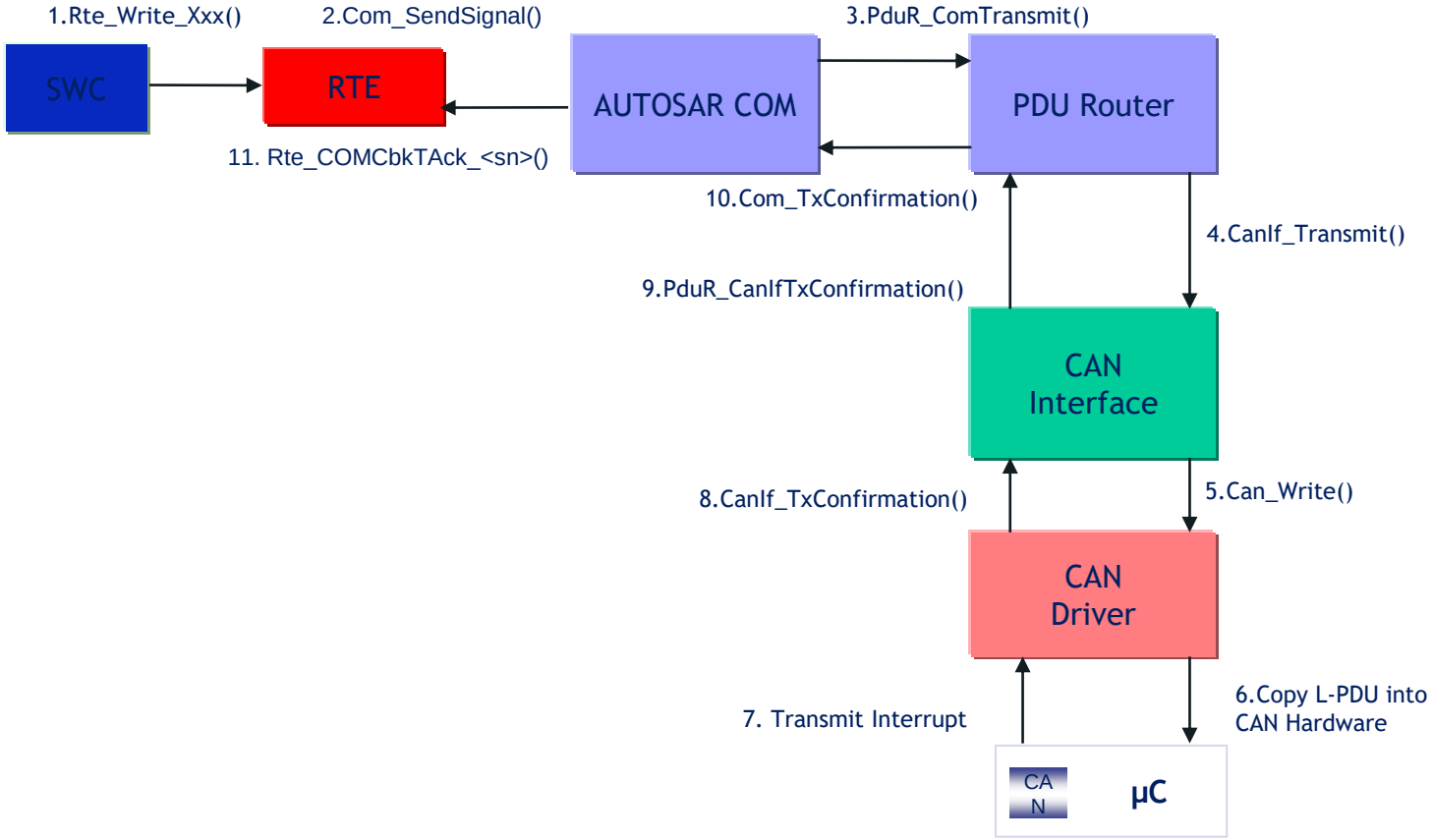
Runtime Environment (RTE)

The RTE uses the capabilities of the AUTOSAR COM module to send and receive signals. In AUTOSAR, the RTE is the higher layer above the AUTOSAR COM module.

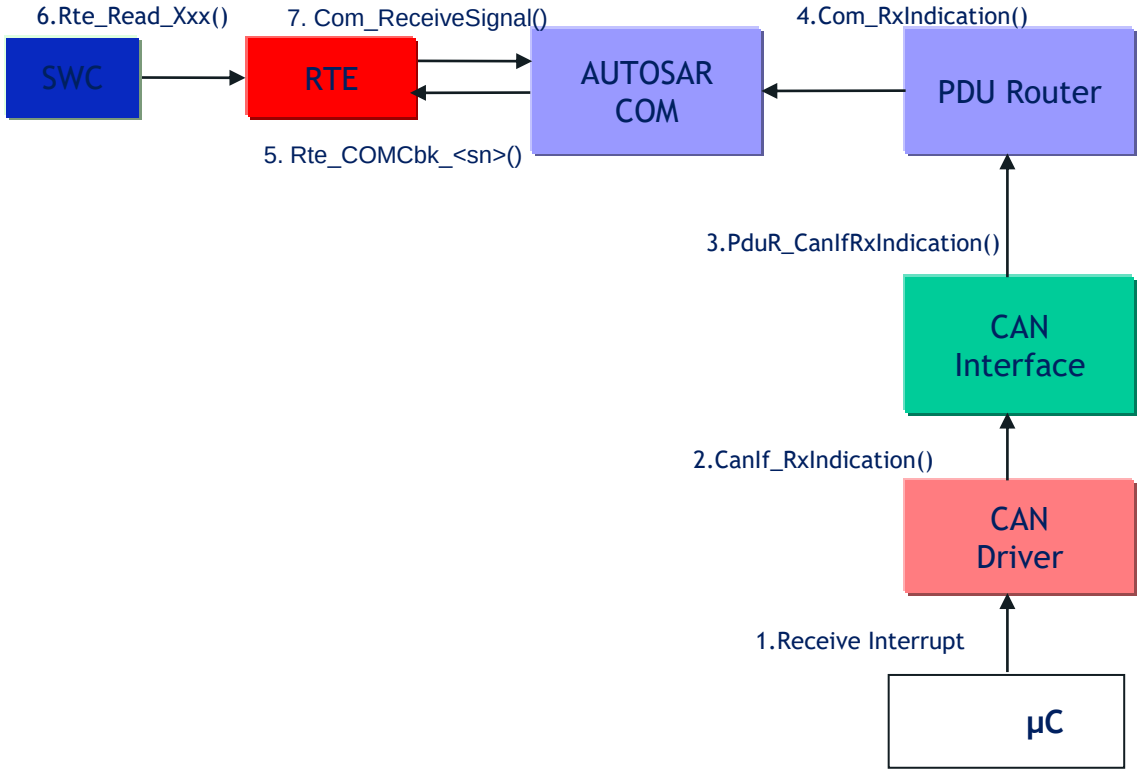
SwCluC

SwCluC establishes inter-Software Cluster connection for exchanging (both providing and consuming) signals and signal groups. The Com users in Applicative Software Clusters can access the signals and signal groups via the SwCluC Com High Proxy. In the Host Software Cluster the SwCluC Com Low Proxy has the interface to the COM module.

Data Flow(Transmit)



Data Flow (Receive)



Start-Up Behavior

- The AUTOSAR COM module shall initialize each I-PDU during execution of Com_Init, firstly byte wise with the **ComTxIPduUnusedAreasDefault** value and then bit wise according to initial values (**ComSignalInitValue**) of the contained signals and the update-bits

- The C initialization code, also known as start-up code, initializes global and static variables with the initial values. It must be executed before any call of an AUTOSAR COM module's service.

- Com_Init shall initialize all internal data that is not yet initialized by the *start-up code e.g. C-structs.*

- By default, all I-PDU groups shall be in the state stopped and they shall not be started automatically by a call to Com_Init

De-Initialization

- The AUTOSAR COM module provides the API function Com_DeInit for de-initialization of the COM layer. This means, after de-initialization of the layer, no communication via the AUTOSAR COM module is possible and all started I-PDU groups are stopped

Endianness Conversion and Sign Extension

The AUTOSAR COM module shall support the following data types:

- boolean
- uint8
- uint16
- uint32
- uint64
- sint8
- sint16
- sint32
- sint64
- uint8[n] (*UINT8_N or UINT8_DYN)*
- float32
- float64

The AUTOSAR COM module shall support endianness conversion for all supported signed and unsigned integer data types

Message reception

- Reception of a message starts with an *indication of the delivery of its containing PDU from* the underlying layer.
- If indication is successful, an I-PDU Callout is called (if configured) and this PDU is copied into the I-PDU.
- After copying the data into the I-PDU further processing is performed separately for each contained message.
- The *Reception Deadline Monitoring takes place ,* Deadline Monitoring can invoke *Message Reception Error notification* when the message reception deadline is missed because the I-PDU that contains the message is not received in time
- Message data is unpacked from the I-PDU.
- The *filtering is applied to the message content. If the message is not filtered out, then the* message data is copied into the receiver message object.

Reception filtering

- Filtering provides a means to discard the received message when certain conditions, set by message filter, are not met for the message value.
- The message filter is a configurable function that filters messages out according to specific algorithms.
- For each message a different filtering condition can be defined through a dedicated algorithm.
- The AUTOSAR COM module shall evaluate each filtering condition to either true or false.

The following attributes are used by the set of filter algorithms :

- *new_value:* current value of the mess*age*
- *old_value*: last value of the message (initialised with the initial value of the message, updated with new_value if the new message value is not filtered out)
- *mask, x, min, max, period, offset:* constant values
- *occurrence:* a count of the number of occurrences of this message

Algorithm Reference	Algorithm	Description
F_Always	True	No filtering is performed so that the message always passes
F_Never	False	The filter removes all messages
F_MaskedNewEqualsX	*(new_value&mask) == x*	Pass messages whose masked value is equal to a specific value
F_MaskedNewDiffersX	*(new_value&mask) != x*	Pass messages whose masked value is not equal to a specific value
F_NewIsEqual	*new_value == old_value*	Pass messages which have not changed
F_NewIsDifferent	*new_value != old_value*	Pass messages which have changed
F_MaskedNewEqualsMaskedOld	*(new_value&mask) == (old_value&mask)*	Pass messages where the masked value has not changed
F_ MaskedNewDiffersMaskedOld	*(new_value&mask) != (old_value&mask)*	Pass messages where the masked value has changed
F_NewIsWithin	*min <= new_value <= max*	Pass a message if its value is within a predefined boundary
F_NewIsOutside	*(min > new_value) OR (new_value > max)*	Pass a message if its value is outside a predefined boundary
F_NewIsGreater	*new_value > old_value*	Pass a message if its value has increased
F_NewIsLessOrEqual	*new_value <= old_value*	Pass a message if its value has not increased
F_NewIsLess	*new_value < old_value*	Pass a message if its value has decreased
F_NewIsGreaterOrEqual	*new_value >= old_value*	Pass a message if its value has not decreased
F_OneEveryN	*occurrence % period == offset*	Pass a message once every N message occurrences. Start: occurrence = 0. Each time the message is received or transmitted, occurrence is incremented by 1 after filtering. Length of occurrence is 8 bit (minimum).

For signals with ComSignalType FLOAT32, FLOAT64, UINT8_N or UINT8_DYN, the AUTOSAR COM module shall only support ComFilterAlgorithm configured to

- *ALWAYS*
- *NEVER*
- *ONE_EVERY_N*

For signals with ComSignalType configured to BOOLEAN, the AUTOSAR COM module shall only support ComFilterAlgorithm configured to:

- *ALWAYS*
- *NEVER*
- *MASKED_NEW_EQUALS_X*
- *MASKED_NEW_DIFFERS_X*
- *MASKED_NEW_DIFFERS_MASKED_OLD*
- *ONE_EVERY_N*

- If the AUTOSAR COM module filters out a signal on receiver side, i.e. filter condition evaluates to false, the AUTOSAR COM module shall discard that signal and shall not process it.

- If the AUTOSAR COM module filters out a group signal on receiver side, i.e. filter condition evaluates to false, the AUTOSAR COM module shall discard the whole signal group and shall not process it.

Message transmission

- Sending a message requires the transfer of the application message to the I-PDU (external communication) and/or the receiving message object (internal communication).
- A message that is transferred can be stored in zero or more message objects for internal receivers and in zero or one I-PDU for external communication.
- Transmission of messages via the underlying layers takes place based on I-PDUs.
- Transmission of I-PDUs may use any of the applicable transmission modes of I-PDUs: Direct, Periodic or Mixed.
- Internal messages do not have transfer properties because the transfer is always executed in the same way. The IL routes internal messages directly to the receiving part of the IL for further processing.

Transfer Properties and I-PDU Transmission Mode

Signals -> transfer properties
I-PDUs -> transmission modes

Transfer Property

Configuration Parameter -> ComTransferProperty

- *TRIGGERED*
- *TRIGGERED_WITHOUT_REPETITION*
- *TRIGGERED_ON_CHANGE*
- TRIGGERED*_ON_CHANGE_WITHOUT_REPETITION*
- *PENDING*

Transmission Modes

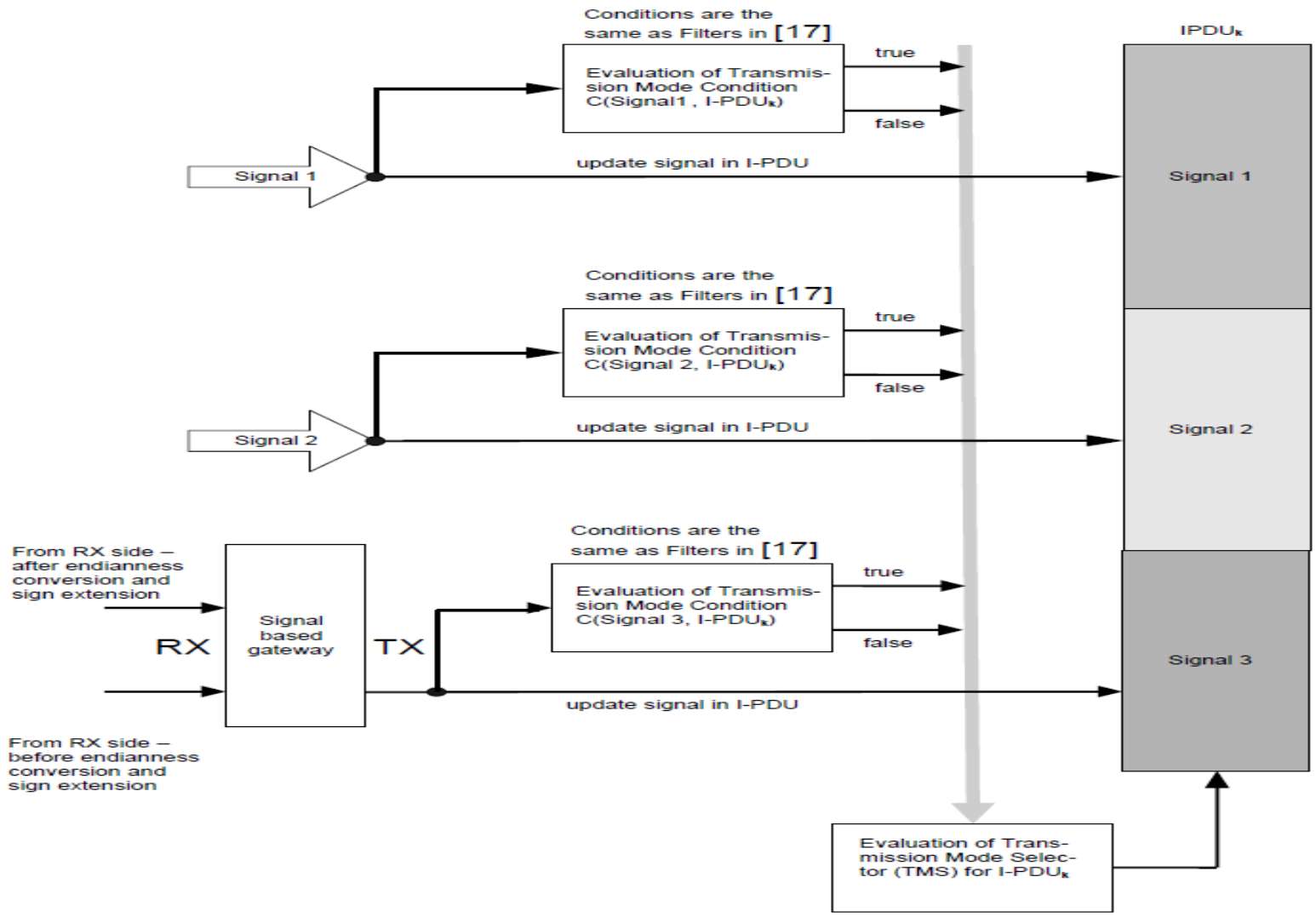
Configuration Parameter -> ComTxModeMode

- *DIRECT*
- *MIXED*
- *PERIODIC*
- *NONE*

TMC -> Transmission Mode Condition
TMS -> Transmission Mode Selection

There are 2 modes:

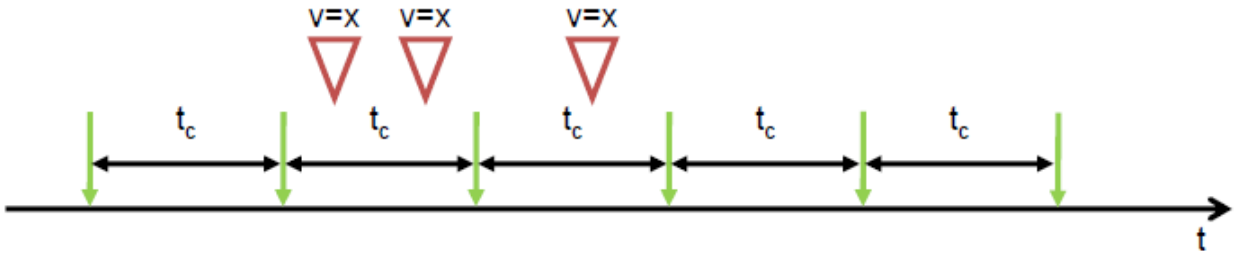
1. True Mode Selection
2. False Mode Selection


Conditions are the same as Filters in [17]
Evaluation of Transmis-sion Mode Condition C(Signal1 , I-PDUk)
true
false
IPDUk
Signal 1
update signal in I-PDU
Signal 1
Conditions are the same as Filters in [17]
Evaluation of Transmis-sion Mode Condition C(Signal 2, I-PDUk)
true
false
Signal 2
update signal in I-PDU
Signal 2
From RX side – after endianness conversion and sign extension
RX
Signal based gateway
TX
Conditions are the same as Filters in [17]
Evaluation of Transmis-sion Mode Condition C(Signal 3, I-PDUk)
true
false
Signal 3
update signal in I-PDU
From RX side – before endianness conversion and sign extension
Evaluation of Trans-mission Mode Selec-tor (TMS) for I-PDUk

Notation	Description
t_c, t_{c1}, t_{c2}	cycle times
t_d	cycle time of N-Times sent signals
t_r	minimum SW reaction time of COM-Layer due to internal main cycle time
V	Value: x stands for an arbitrary value/ value range, a…w for specific values / value ranges, defined by the user, with a <> b, *range a* is disjoint from *range b*.
	request from RTE to the COM-Layer
	request from COM-Layer to PDU Router
	potential but skipped request from COM-Layer to PDU Router (e.g. because of a new send request by the RTE or delayed due to minimum delay time)
dt	minimum delay time: minimum distance between two requests to PDU Router
without TMS switch	without switching of the TMS (see 7.3.3.2) from *true* to *false* or vice versa
with TMS switch	with switching of the TMS (see 7.3.3.2) from *true* to *false* or vice versa (from TM 1 to TM 2); one TM is named before the "+" and one behind in the description

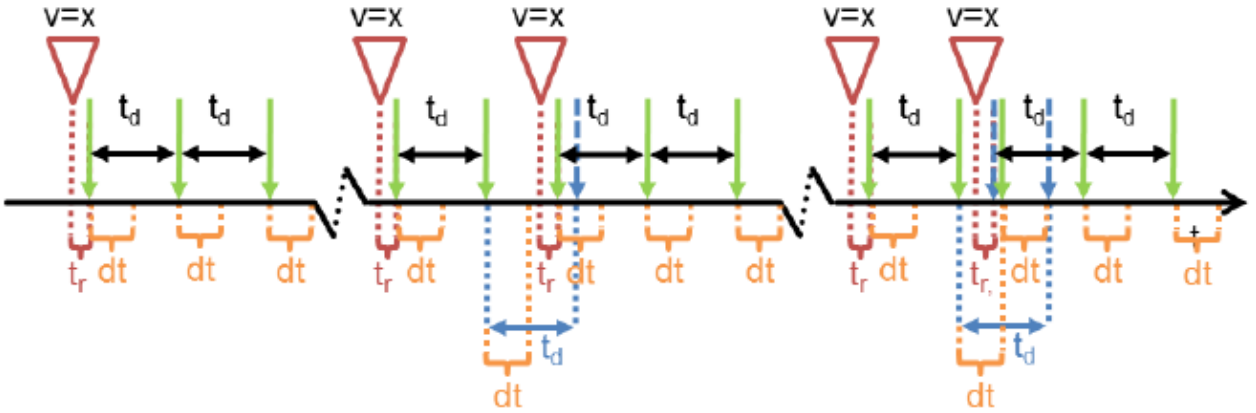
Table 9: Legend for use case diagrams.

Use case 1



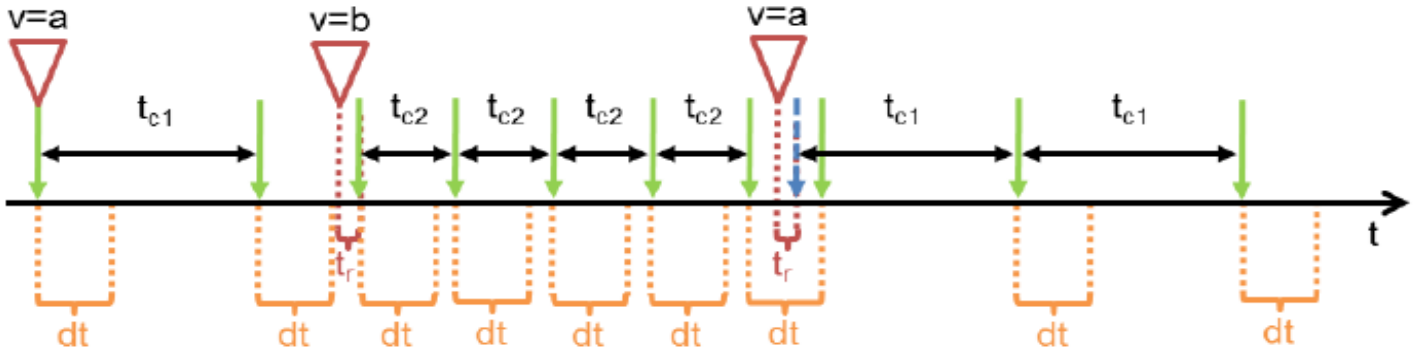
ComSignal	
ComTransferProperty	PENDING or TRIGGERED (TRIGGERED has no influence)
ComFilter	
ComFilterAlgorithm	ALWAYS
ComIPdu	
ComIPduDirection	SEND
ComTxModeTrue	
ComTxModeTimePeriod	tc
ComTxModeMode	PERIODIC
ComTxIPdu	
ComMinimumDelayTime	0 or omitted

Use case 2



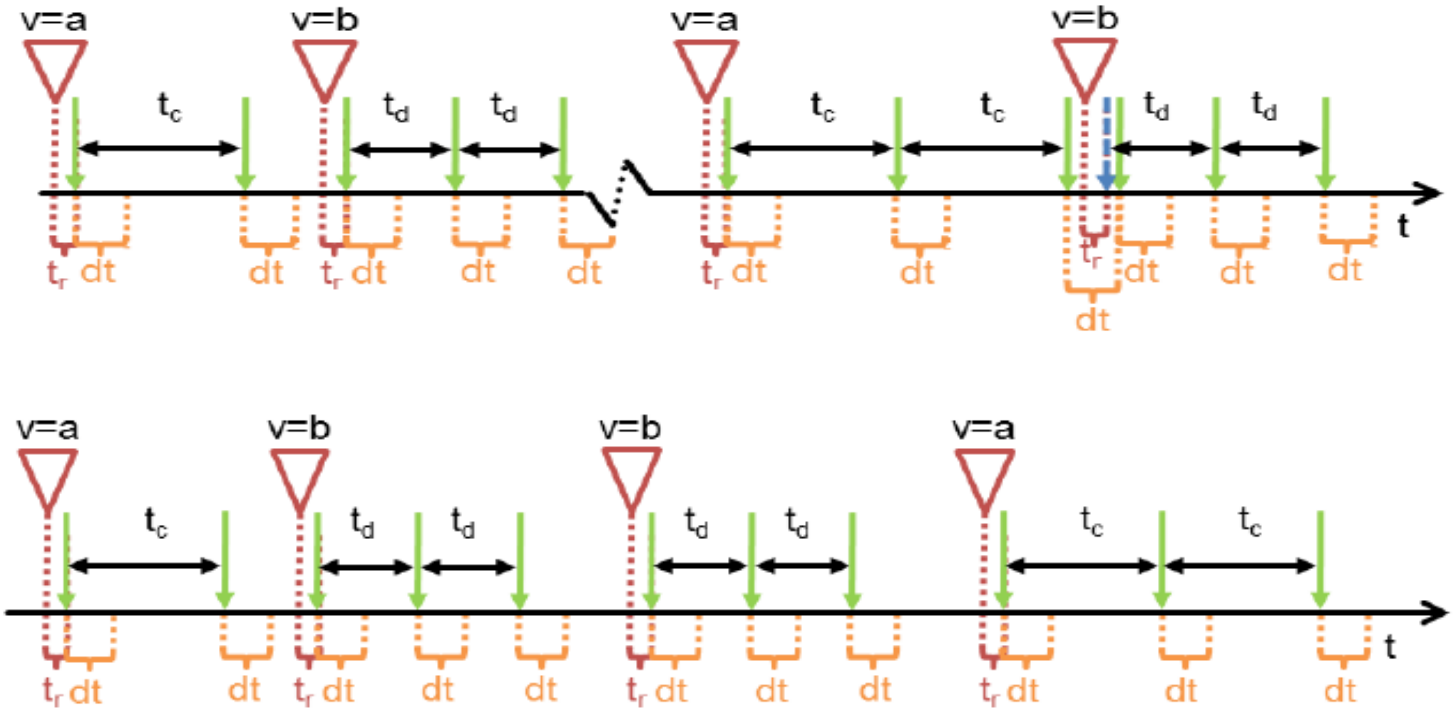
ComGeneral	
ComEnableMDTForCyclicTransmission	true
ComSignal	
ComTransferProperty	TRIGGERED
ComFilter	
ComFilterAlgorithm	ALWAYS
ComIPdu	
ComIPduDirection	SEND
ComTxModeTrue	
ComTxModeMode	DIRECT
ComTxModeNumberOfRepetitions	2
ComTxModeRepetitionPeriod	t_d
ComTxIPdu	
ComMinimumDelayTime	dt

Use case 3



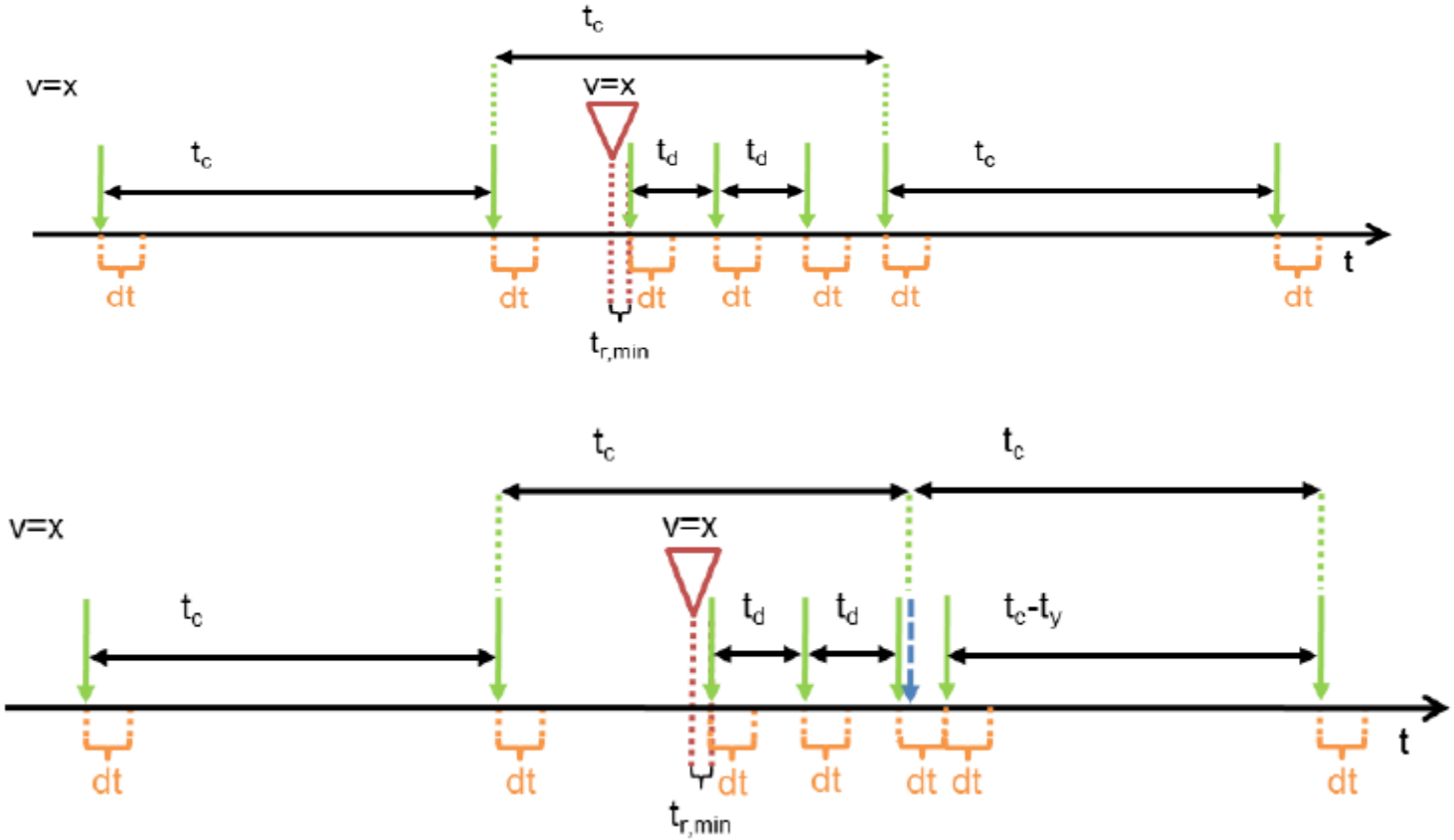
ComGeneral	
ComEnableMDTForCyclicTransmission	true
ComSignal	
ComTransferProperty	PENDING or TRIGGERED (TRIGGERED has no influence)
ComFilter	
ComFilterAlgorithm	any filter mapping value a to true and value b to false
ComIPdu	
ComIPduDirection	SEND
ComTxModeTrue	
ComTxModeMode	PERIODIC
ComTxModeTimePeriod	t_{c1}
ComTxModeFalse	
ComTxModeMode	PERIODIC
ComTxModeTimePeriod	t_{c2}
ComTxIPdu	
ComMinimumDelayTime	dt

Use case 4


v=a
v=b
v=a
v=b
t_c
t_d
t_d
t_c
t_c
t_d
t_d
t
t_r dt
dt
t_r dt
dt
dt
t_r dt
dt
t_r
dt
dt
dt
dt
v=a
v=b
v=b
v=a
t_c
t_d
t_d
t_d
t_d
t_c
t_c
t
t_r dt
dt
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t_r dt
dt
dt
t_r dt
dt
dt

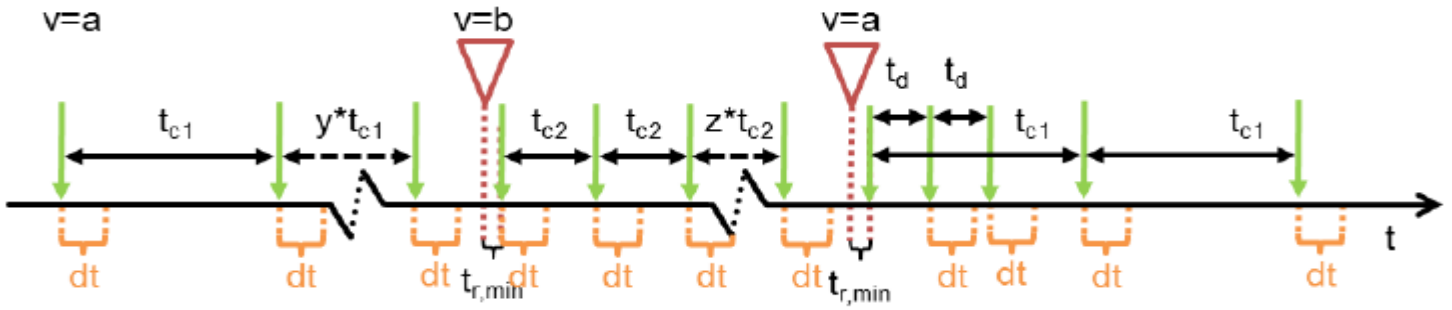
ComGeneral	
ComEnableMDTForCyclicTransmission	True
ComSignal	
ComTransferProperty	TRIGGERED
ComFilter	
ComFilterAlgorithm	any filter mapping value a to true and value b to false
ComIPdu	
ComIPduDirection	SEND
ComTxModeTrue	
ComTxModeMode	PERIODIC
ComTxModeTimePeriod	t_c
ComTxModeFalse	
ComTxModeMode	DIRECT
ComTxModeNumberOfRepetitions	2
ComTxModeRepetitionPeriod	t_d
ComTxIPdu	
ComMinimumDelayTime	dt

Use case 5


t_c
v=x
v=x
t_c
t_d
t_d
t_c
t
dt
dt
dt
dt
dt
dt
dt
$t_{r,min}$
t_c
t_c
v=x
v=x
t_c
t_d
t_d
t_c-t_y
t
dt
dt
dt
dt
dt
dt
dt
$t_{r,min}$

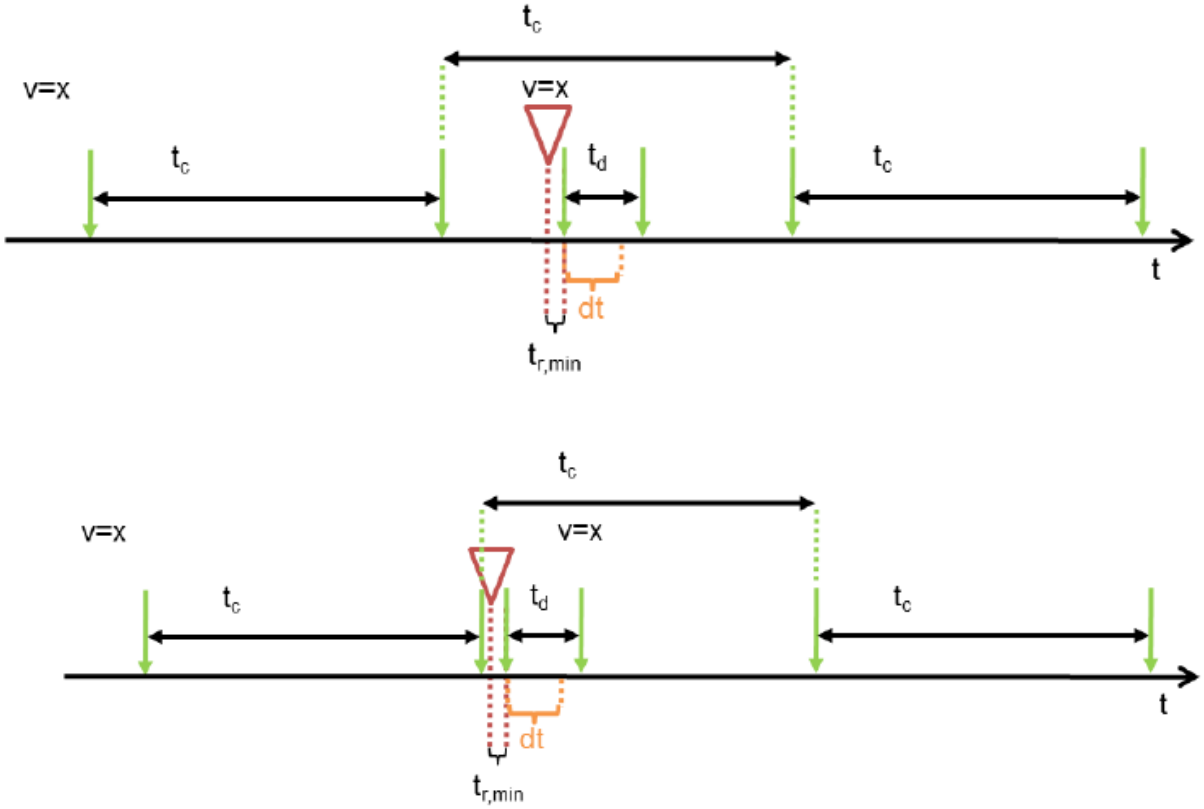
ComGeneral	
ComEnableMDTForCyclicTransmission	true
ComSignal	
ComTransferProperty	TRIGGERED
ComFilter	
ComFilterAlgorithm	ALWAYS
ComIPdu	
ComIPduDirection	SEND
ComTxModeTrue	
ComTxModeMode	MIXED
ComTxModeTimePeriod	t_c
ComTxModeNumberOfRepetitions	2
ComTxModeRepetitionPeriod	t_d
ComTxIPdu	
ComMinimumDelayTime	dt

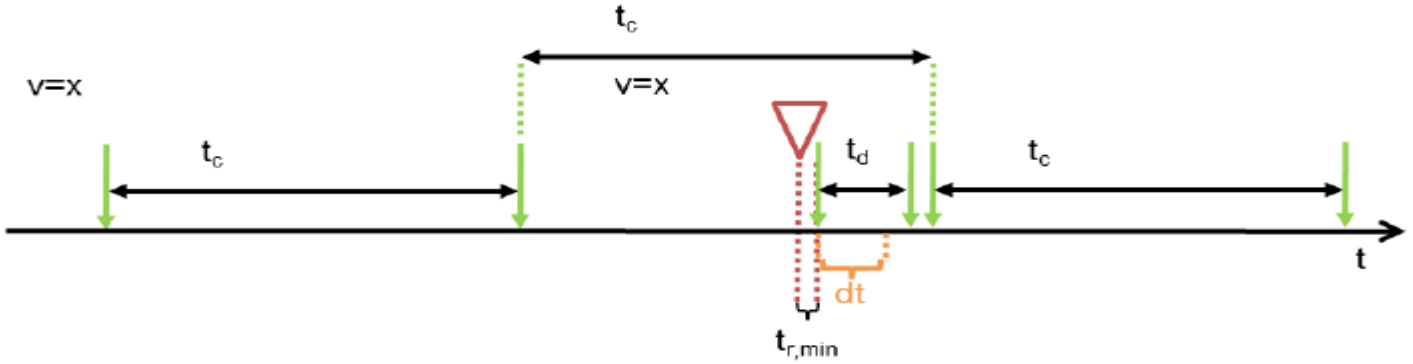
Use case 6



ComGeneral	
ComEnableMDTForCyclicTransmission	true
ComSignal	
ComTransferProperty	TRIGGERED
ComFilter	
ComFilterAlgorithm	any filter mapping value a to true and value b to false
ComIPdu	
ComIPduDirection	SEND
ComTxModeTrue	
ComTxModeMode	MIXED
ComTxModeTimePeriod	t_{c1}
ComTxModeNumberOfRepetitions	2
ComTxModeRepetitionPeriod	t_d
ComTxModeFalse	
ComTxModeMode	PERIODIC
ComTxModeTimePeriod	t_{c2}
ComTxIPdu	
ComMinimumDelayTime	dt

Use case 7





ComGeneral	
ComEnableMDTForCyclicTransmission	False
ComSignal	
ComTransferProperty	TRIGGERED
ComFilter	
ComFilterAlgorithm	ALWAYS
ComIPdu	
ComIPduDirection	SEND
ComTxModeTrue	
ComTxModeMode	MIXED
ComTxModeTimePeriod	t_c
ComTxModeNumberOfRepetitions	1
ComTxModeRepetitionPeriod	t_d
ComTxIPdu	
ComMinimumDelayTime	dt

Signal Invalidation

Com provides possibility for the sender to indicate that it is not able to provide a valid value for a corresponding signal.
example -> in case a sensor is faulty.

Configuration parameter -> ComSignalDataInvalidValue

Transmission of an Invalidated Signal

Call to Com_InvalidateSignal -> Com_SendSignal (ComSignalDataInvalidValue)
data invalid value -> used as current value for filters and TMS.

Reception of an Invalidated Signal

If ComDataInvalidAction -> *NOTIFY*

Com will notify users (e.g. RTE, SwCluC) via the configured ComInvalidNotification function and no other signal processing like filtering or the normal signal indication shall take place

If ComDataInvalidAction -> *REPLACE*

Com will replace the signal's value by its configured ComSignalInitValue

Handling of I-PDU Groups

- An I-PDU Group is an arbitrary collection of I-PDUs of the same direction (i.e. send or receive) in the COM module.
- **For an I-PDU group the following rules apply**:

i. An I-PDU can belong to any I-PDU group.
ii. An I-PDU is active (started) if it belong to a started I-PDU group or it does not belong to any I-PDU group and it was started implicitly.
iii. The maximum number of I-PDU groups is pre-compile configurable.
iv. I-PDU groups may contain only I-PDUs of the same direction (i.e. send or receive)

- **ComSupportedIpduGroups** -> Maximum number of supported I-PDU groups
- The nesting of I-PDU groups is purely conceptual and must be resolved by an appropriate configuration and/ or usage.
- Complete state handling of I-PDU groups is done out-side of the AUTOSAR COM module, e.g. within the BswM.
- Module managing the I-PDU group states consistently starts or stops the I-PDU groups via **Com_IpduGroupStart** and **Com_IpduGroupStop**
- If an I-PDU is not assigned to any I-PDU group, the AUTOSAR COM shall start this I-PDU within **Com_Init** . Since the AUTOSAR COM never stops such an I-PDU implicitly and it is not assigned to any I-PDU group, such an I-PDU can never be stopped

Starting of I-PDU Groups

- By default all I-PDU groups are stopped

- **Com_IpduGroupStart** -> Starts IPDU Group, COM module shall permit to transmit/ receive its signals and signal groups

- If an I-PDU is started by Com_IpduGroupStart, COM module shall always initialize the following attributes of this I-PDU:
  - ComMinimumDelayTime of I-PDUs in transmission mode *DIRECT or MIXED*
  - Restart all reception deadline monitoring timers for all signals with a non-zero configured ComFirstTimeout
  - Cancel all transmission deadline monitoring timers and use ComFirstTimeout (if configured) as value when transmission timer is started the first time after the I-PDU activation
  - all included update-bits shall be cleared
  - reset *OCCURRENCE of filters with ComFilterAlgorithm ONE_EVERY_N*

- If an I-PDU is started by Com_IpduGroupStart with parameter Initialize set to true, then COM will initialise
  - the data of the I-PDU
  - the shadow buffers of included signal groups
  - old_value of the filtering mechanisms for each signal to the ComSignalInit-Value
  - ComTxModeTimePeriod and ComTxModeTimeOffset of I-PDUs in Periodic or MIXED transmission mode

Behavior of started I-PDUs	
Receiver side (RX)	***Transmitter side (TX)***
• Reinitialize timeouts if Initialize==true (ComSignalFirstTimeout, ComSignal-Timeout) • Normal reaction on Com_RxIndication/ Com_TpRxIndication • Normal reaction on Com_ReceiveSignal, and Com_ReceiveSignalGroup	• Normal reaction on Com_InvalidateSignal, Com_SendSignal, and Com_SendSignalGroup • No transmission timeout notification until next send • Normal reaction on Com_TxConfirmation/ Com_TpTxConfirmation • Normal reaction on Com_TriggerTransmit
	For periodic (TX)
	Start at 0

Stopping of I-PDU Groups

Com_IpduGroupStop -> stops an I-PDU group

- In order to disable the transmission of an I-PDU, all I-PDU groups containing this I-PDU needs to be stopped.
- By a call to the functions: Com_SendSignal, Com_Send-SignalGroup, or Com_InvalidateSignal, the AUTOSAR COM module shall update the values of its internal buffers even for stopped I-PDUs
- If a signal written to a stopped I-PDU would trigger the transmission of this I-PDU if it were not stopped, then this trigger is not stored. After re-starting the corresponding I-PDU group, such an old trigger does not lead to an immediate transmission of the I-PDU.

If an I-PDU is stopped by Com_IpduGroupStop:

- The AUTOSAR COM module shall cancel any outstanding transmission requests for this I-PDU.
- The AUTOSAR COM module shall ignore any transmit confirmations for a stopped I-PDU
- The function Com_Trigger-Transmit returns E_NOT_OK.
- COM module shall disable its reception processing.
- For large I-PDU the COM module shall stop the reception process and ignore the partly received I-PDU.

- COM module shall cancel its deadline monitoring
- COM module shall immediately invoke the configured ComErrorNotification for outstanding not confirmed transmitted signals/ signal groups of the stopped I-PDU.
- If a large I-PDU is stopped while its transmission is already in progress, the AUTOSAR COM module shall stop the transmission process immediately.

Behavior of stopped I-PDUs	
Receiver side (RX)	***Transmitter side (TX)***
• Disable RX deadline monitoring • No action on a Com_RxIndication to users (e.g. RTE, SwCluC), no storing of the I-PDU • Return code COM_SERVICE_NOT_AVAILABLE on Com_ReceiveSignal and Com_ReceiveSignalGroup and the last known value (or init value) is given back as data	• Disable sending • Disable TX deadline monitoring • Ignore Com_TxConfirmation: • On a call of Com_SendSignal, Com_SendSignalGroup, Com_InvalidateSignal the values in the AUTOSAR COM module's internal buffers are still up-dated but the return code COM_SERVICE_NOT_AVAILABLE is returned • Outstanding transmission request (e.g. N-Times) will be cancelled • Return code E_NOT_OK on Com_TriggerTransmit
	For periodic (TX)
	Do not send any more

Behavior of stopped I-PDUs	
Receiver side (RX)	***Transmitter side (TX)***
• Disable RX deadline monitoring • No action on a Com_RxIndication to users (e.g. RTE, SwCluC), no storing of the I-PDU • Return code COM_SERVICE_NOT_AVAILABLE on Com_ReceiveSignal and Com_ReceiveSignalGroup and the last known value (or init value) is given back as data	• Disable sending • Disable TX deadline monitoring • Ignore Com_TxConfirmation: • On a call of Com_SendSignal, Com_SendSignalGroup, Com_InvalidateSignal the values in the AUTOSAR COM module's internal buffers are still up-dated but the return code COM_SERVICE_NOT_AVAILABLE is returned • Outstanding transmission request (e.g. N-Times) will be cancelled • Return code E_NOT_OK on Com_TriggerTransmit
	For periodic (TX)
	Do not send any more

Behavior of started I-PDUs	
Receiver side (RX)	***Transmitter side (TX)***
• Reinitialize timeouts if Initialize==true (ComSignalFirstTimeout, ComSignal-Timeout) • Normal reaction on Com_RxIndication/ Com_TpRxIndication • Normal reaction on Com_ReceiveSignal, and Com_ReceiveSignalGroup	• Normal reaction on Com_InvalidateSignal, Com_SendSignal, and Com_SendSignalGroup • No transmission timeout notification until next send • Normal reaction on Com_TxConfirmation/ Com_TpTxConfirmation • Normal reaction on Com_TriggerTransmit
	For periodic (TX)
	Start at 0

Signal Indication (Unpacking of I-PDUs)

Configuration parameter -> ComIPduSignalProcessing.

configurable signal indication modes -> *IMMEDIATE and DEFERRED*

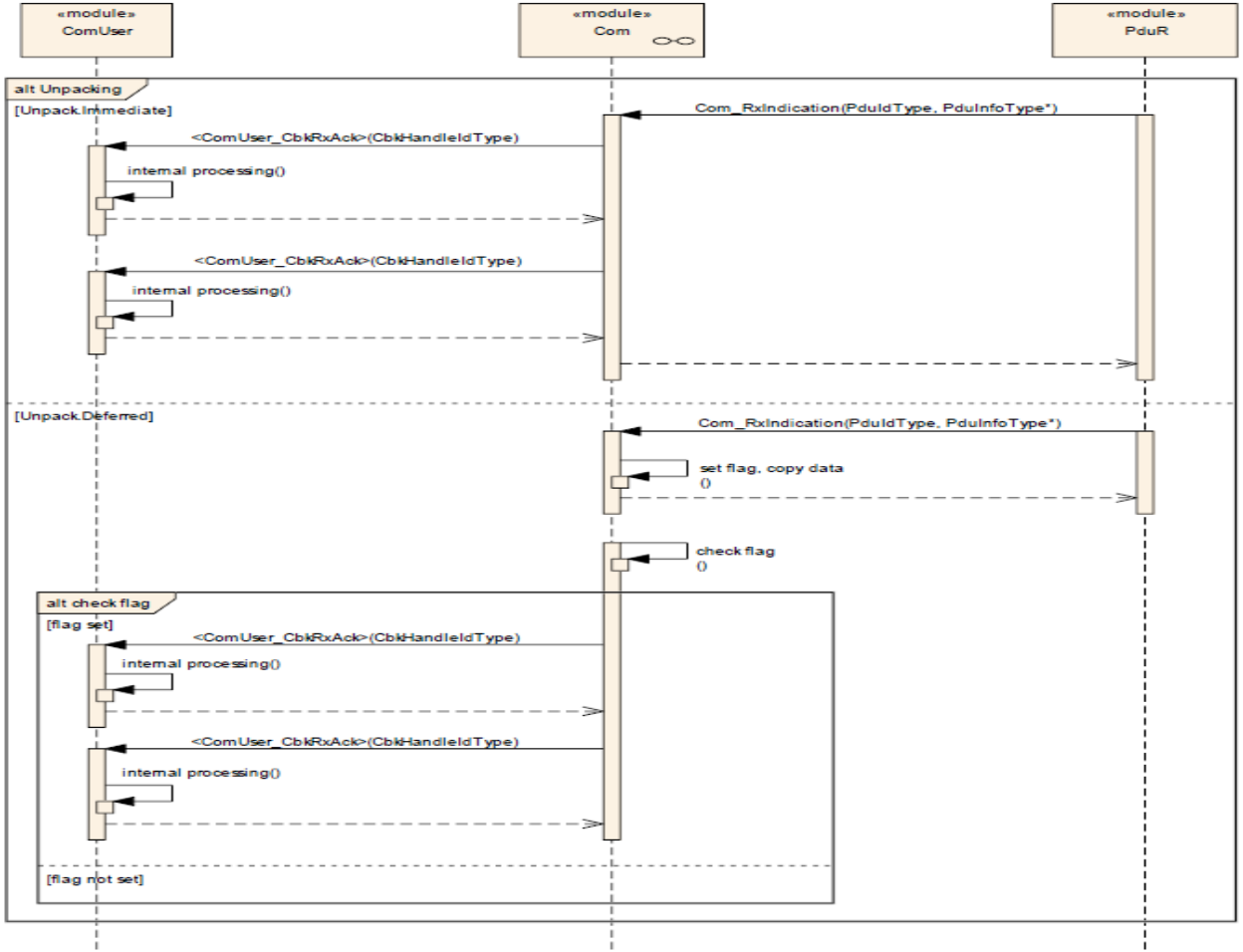
IMMEDIATE

- COM module shall invoke the configured ComNotifications for the included signals and signal groups within the Com_RxIndication, or Com_TpRxIndication function respectively.

DEFERRED

- COM module shall first copy the relevant data (e.g. configured ComSignal) of the received I-PDU within the Com_RxIndication function or the related TP reception functions respectively from the PduR into COM.

- COM module shall invoke the configured ComNotifications for the included signals and signal groups asynchronously during the next call to Com_MainFunctionRx


«module»
ComUser
«module»
Com
«module»
PduR
alt Unpacking
[Unpack.Immediate]
Com_RxIndication(PduIdType, PduInfoType*)
<ComUser_CbkRxAck>(CbkHandleIdType)
internal processing()
<ComUser_CbkRxAck>(CbkHandleIdType)
internal processing()
[Unpack.Deferred]
Com_RxIndication(PduIdType, PduInfoType*)
set flag, copy data
()
check flag
()
alt check flag
[flag set]
<ComUser_CbkRxAck>(CbkHandleIdType)
internal processing()
<ComUser_CbkRxAck>(CbkHandleIdType)
internal processing()
[flag not set]

Minimum Delay Timer (MDT)

MDT counts minimum delay time, which is a minimum distance in time between two requests from COM to PDU Router, for which the successive request should wait.

When an I-PDU is started, the MDT is re-initialized. Therefore, the MDT can be violated by stopping and starting I-PDUs rapidly.

No minimum delay time monitoring shall take place, if **ComMinimumDelayTime** is omitted or configured to 0.

If **ComEnableMDTForCyclicTransmission** -> false, MDT shall not be (re)started in :

- I-PDUs with ComTxModeMode PERIODIC
- the cyclic part of I-PDUs with ComTxModeMode MIXED
- repeated transmissions for I-PDUs in case ComTxModeNumberOfRepetitions > 0

If **ComMinimumDelayTime** > 0 and **ComRetryFailedTransmitRequests** -> *FALSE*

- COM module shall load and start the minimum delay time counter upon transmission of that I-PDU to the PDU Router via PduR_ComTransmit

If **ComMinimumDelayTime** > 0 and **ComRetryFailedTransmitRequests** -> *TRUE*

- COM module shall load and start the minimum delay time counter upon transmission of that I-PDU to the PDU Router via PduR_ComTransmit only in case PduR_ComTransmit returns E_OK.

- The running minimum delay timer is reloaded upon the reception of a successful TX-confirmation of that I-PDU, unless the transmission was already delayed longer than ComMinimumDelayTime at the reception of the confirmation.

- Com_TriggerTransmit does not interfere with the MDT.

- MDT is reset if the transmission deadline monitoring timer expires.

- Starting an I-PDU group resets the MDT of the included I-PDUs.

Reception Deadline Monitoring

- Reception Deadline Monitoring can be used to verify on the receiver side that periodic messages are received within the allowed time frame. This mechanism is configured per message and is performed by monitoring the reception of the I-PDU that contains the message.

- The deadline monitoring mechanism monitors that a periodic message is received within a given time interval

- In case of an Rx-timeout, the **ComRxDataTimeoutAction** parameter determines whether the AUTOSAR COM module replaces the signal/ signal group value with the initial value or substitute value or maintains the last received value.

- If **ComRxDataTimeoutAction -> *REPLACE*** *:* COM module shall replace the signal's value by its ComSignalInitValue when the reception deadline monitoring timer of a signal expires.

- If **ComRxDataTimeoutAction -> SUBSTITUTE** : COM module shall replace the signal's value by its ComTimeoutSubstitutionValue when the reception deadline monitoring timer of a signal expires

- If **ComRxDataTimeoutAction -> NONE** : COM module shall not replace the signal or the signals of the signal groups

Enable / Disable Reception Deadline Monitoring

- When reception deadline monitoring of an I-PDU is disabled and the timer expires, no error indication will be given to the users (e.g. RTE, SwCluC).
- Disabling reception deadline monitoring does not stop the reception of an I-PDU.
- If the reception deadline monitoring of an I-PDU is enabled by Com_EnableReceptionDM, the COM module shall set the reception deadline monitoring timer for the included signals and signal groups to the configured ComFirstTimeout value.
- Timeout notifications of deadline monitoring expiry are notified to the users (e.g. RTE, SwCluC) for signals or signal groups with a configured ComTimeoutNotification.

Transmission Deadline Monitoring

- Deadline Monitoring on the sender side can be used to verify that transmission requests (periodic or otherwise) are followed by transmissions on the network within a given time frame.

- The general idea of the AUTOSAR COM transmission deadline monitoring is to supervise the lower layers and the bus but not the COM module itself. Hence, the transmission monitoring timer is generally started, when the COM module sends an I-PDU to the lower layer. This is independent of a transmission mode, a transfer property or update-bits. In generally the transmission monitoring timer is not re-started or reset if it is currently running.

- The transmission deadline monitoring timer shall be started with the configured ComFirstTimeout value if the timer is started for the first time after a (re-)start of the transmission deadline monitoring service for this I-PDU, otherwise the timer shall be started with ComTimeout value

Notification handling

The notifications are configurable via ComNotification

For each ComNotification that is configured for a signal or sig-nal group, the AUTOSAR COM shall indicate each successful transmission/reception via the configured notifications.

For example, for periodic transmissions each successful periodic transmission is notified.

Note that for n-times transmission multiple transmissions on the bus are necessary before it is considered as successful

Signal Groups - Complex Data Types

- To support the AUTOSAR concept of complex data types the AUTOSAR COM module provides signal groups.
- The AUTOSAR COM module transmits and receives signal groups consistently to provide the necessary consistency for complex data types.
- Signal groups can be configured statically. For each signal group a symbolic name can be configured.
- The AUTOSAR COM module achieves the consistency of a signal group by means of a shadow buffer mechanism. the user (e.g. RTE or SwCluC) accesses the group signals in the shadow buffer.
- If the shadow buffer needs to be synchronized with the I-PDU, the users can trigger this explicitly with Com_SendSignalGroup or Com_ReceiveSignalGroup. The synchronization is performed atomically.
- An alternative approach to signal group consistency is to enable the signal group array API Com_SendSignalGroupArray and Com_ReceiveSignalGroupArray

Initialization

By a call to Com_Init, the AUTOSAR COM module shall initialize the shadow buffer of a signal group on sender-side

Transmission

If Com_SendSignal or Com_InvalidateSignal is called for a signal that belongs to a signal group, then the AUTOSAR COM will only update the shadow buffer of this signal group.

There is no need for any further I-PDU processing like TMS evaluation, unless the I-PDU contents changed

If Com_SendSignalGroup is called for the signal group, the AUTOSAR COM module shall copy the shadow buffer atomically to the I-PDU buffer.

```
/* copy a to shadow buffer */
Com_SendSignal (signal_a, &a);
/* copy b to shadow buffer */
Com_SendSignal (signal_b, &b);
/* copy shadow buffer to I-PDU */
Com_SendSignalGroup (group_x);
```

Reception

A group signal can be received from the shadow buffer by calling the function Com_ReceiveSignal after the signal group data has been copied to the shadow buffer by Com_ReceiveSignalGroup.

If Com_ReceiveSignalGroup is called for a signal group, data is atomically copied from the I-PDU buffer to the shadow buffer.

```
/* copy I-PDU to shadow buffer */
Com_ReceiveSignalGroup (group_x);

/* copy a from shadow buffer */
Com_ReceiveSignal (signal_a, &a);

/* copy b from shadow buffer */
Com_ReceiveSignal (signal_b, &b);
```

Notifications

Notification functions are only configurable for signal groups but not for individual group signals

Attributes of a Signal Group

Attribute	*Per group signal*	*Per signal group*
Update-bit	No	Yes, associated on the whole group (see Chapter 7.9)
Signal Notification (sender side)	No	Yes
Signal Notification (receiver side)	No	Yes
Error Notification (sender side)	No	Yes
Timeout Notification (sender side)	No	Yes
Timeout Notification (receiver side)	No	Yes
Invalid Notification (receiver side)	No	Yes
Data access (receiver side)	Yes	Yes, see SWS_Com_00201
Data access (sender side)	Yes	Yes, see SWS_Com_00200, SWS_Com_00557
Data Filtering (receiver side)	Yes, see Chapter 7.2.4	No
Data Filtering (sender side)	No	No
TMS on sender side	Each signal, according to TMS selection definition. (see Chapter 7.3.3.2)	No

UINT8 array based access to signal groups

An alternative approach to signal group consistency is to enable the signal group array APIs **Com_SendSignalGroupArray** and **Com_ReceiveSignalGroupArray**

configuration parameter -> **ComSignalGroupArrayAccess**.

The serialization of the composite data is then done outside of the COM module but resulting in the same layout as COM would use to place the group signals in the I-PDU.

In this approach**, no shadow buffer is used**. The serialized data is directly provided to the COM module as a uint8 array representation of the signal group.

Following APIs is not supported for that signal group:

- Com_SendSignal and Com_ReceiveSignal for any of the group signals of that signal group
- Com_SendSignalGroup and Com_ReceiveSignalGroup for that signal group

certain APIs are not supported for signal groups with array access.

Following filters are supported for reception filtering and Transmission Mode Selection:

- *ALWAYS*
- *NEVER*
- *MASKED_NEW_EQUALS_X*
- *MASKED_NEW_DIFFERS_X*
- *MASKED_NEW_DIFFERS_MASKED_OLD*
- *ONE_EVERY_N*

Basic functionality

The start position of the uint8-array representation of the communication data shall be the byte offset of the group signal in this signal group with **the smallest ComBitPosition**.

The end position of the uint8-array representation of the communication data shall be the byte offset of the last byte occupied by the group signal in this signal group with **the highest ComBitPosition**.

Initialization

By a call to Com_Init, the AUTOSAR COM module shall initial-ize the respective I-PDU buffer section of a signal group, that has ComSignalGroupArrayAccess configured true, based on the ComSig-nalInitValue of all included ComGroupSignals

Transmission

When Com_SendSignalGroupArray is called a data pointer to the uint8-array representation of the signal group is provided. The AUTOSAR COM module shall copy the data into the I-PDU based on the start position and defined size

Reception

When Com_ReceiveSignalGroupArray is called a data pointer to the uint8-array representation of the signal group is provided. The AUTOSAR COM module shall copy the data to the data pointer position of the I-PDU based on the start position and defined size

Large Data Types

- A large signal is a signal that is too large to fit into a single L-PDU of the underlying communication protocol.

  Ex: large signal on CAN would exceed 8 bytes or 64 bytes for CAN FD.

- Large signals in AUTOSAR require configuring a large I-PDU that will be transmitted via the transport protocol of the underlying bus.

- The I-PDU length and hence the signal size will is limited according to the bus-specific transport protocol

- Since the AUTOSAR COM module is not aware of the underlying bus properties, it has to be configured if an I-PDU can be transmitted within a single L-PDU, e.g. one CAN frame, or if it needs to be transmitted via TP,

Transmission of Large Signals/ I-PDUs

- COM module sends large I-PDUs via the generic PduR APIs for up-per layer modules that use transport protocol**.** The call-sequence is:

- **PduR_ComTransmit:** for initiating the send request
- **one or more callbacks to Com_CopyTxData:** for copying the data
- **callback to Com_TpTxConfirmation:** for confirming the transmission

- In case a large I-PDU is currently transmitted and the same I-PDU is triggered for transmission again, the AUTOSAR COM shall skip the additionally send request and report the runtime error COM_E_SKIPPED_-TRANSMISSION.

Reception of Large Signals/ I-PDUs

COM module receives large I-PDUs via the generic PDU Router's APIs for upper layer modules that use transport protocol the call-sequence is:

- **call to Com_StartOfReception:** for initiating the RX request
- **one or more callbacks to Com_CopyRxData:** copying the receive data to the AUTOSAR COM module's internal receive buffer
- **call to Com_TpRxIndication:** indicating the correct, or incorrect, end of the re-ception process

Dynamic Length Signals

- A dynamic length signal is a signal which length can vary at run-time.
- The **maximal length** of a dynamic length signal needs to be **specified at configuration time**. A dynamic length signal has to **be transmitted i**n a **dynamic length I-PDU**.
- For dynamic length signals, the AUTOSAR COM module shall only support the **ComSignalType *UINT8_DYN***.
- It is allowed to configure an update-bit for a dynamic length signal.
  In this case, the update-bit must be located in front of the dynamic length signal
- Dynamic length signals are not restricted to be sent via transport protocol.
- However, the maximum length of the I-PDU containing the dynamic length signal must respect the bus-specific restrictions.

Transmission of Dynamic Length Signals/ I-PDUs

- For sending a dynamic length signal, the user (e.g. RTE or SwCluC) needs to **specify the actual length** at the send call.
- The corresponding API is **Com_SendDynSignal**
- At a call to **Com_SendDynSignal**, the AUTOSAR COM module shall **set the length of the corresponding dynamic length I-PDU to the smallest length.**
- The length of the dynamic signal shall be equal to the value of input parameter "**Length**" passed in the API **Com_SendDynSignal**.
- When the AUTOSAR COM module invokes **PduR_ComTransmit** for a dynamic length I-PDU, the SduLength of the PduInfoType shall be set according the last update (or init value if not yet sent) of the included dynamic length signal.

Reception of Dynamic Length Signals/ I-PDUs

- For receiving a dynamic length signal, the users (e.g. RTE, SwCluC) need to be in-formed about the actual size.
- At reception of a dynamic length I-PDU, the AUTOSAR COM module shall calculate the length of the contained dynamic length signal, by:

<dynamic signal length in bytes> = <received I-PDU length in bytes> - <ComBitPosition of dynamic length signal> / 8

Interface between AUTOSAR COM Module and the PDU Router

- The only lower layer that the AUTOSAR COM module interfaces to is the PDU Router.
- The AUTOSAR COM module uses the PDU Router in two different modes, depending on the type of the I-PDU it will be transported with the bus-specific transport protocol or without.
- If an I-PDU is supposed to be sent via TP, is configured per configuration parameter ComIPduType. The AUTOSAR COM module shall send out I-PDUs by a calling the PduR_ComTransmit function
- **ComIPduType** -> **NORMAL or TP**

API	*normal I-PDUs*	*TP I-PDUs*
PduR_ComTransmit	X	X
Com_TxConfirmation	X	-
Com_RxIndication	X	-
Com_TriggerTransmit	X	-
Com_TpTxConfirmation	-	X
Com_TpRxIndication	-	X
Com_StartOfReception	-	X
Com_CopyRxData	-	X
Com_CopyTxData	-	X

Table 6: API to PDU Router with respect to ComIPduType

Update-Bits

- To enable the receiver of a signal/ signal group to identify whether the sender has updated the data in this signal/ signal group before sending, the AUTOSAR COM module supports *update-bits.*
- The update-bits indicate whether the user (e.g. RTE or SwCluC) on sender-side has updated a signal value, before the I-PDU, containing that signal, was transmitted to the PDU Router.
- Update-bits are not allowed if transmission mode *DIRECT is used with ComTxModeNumberOfRepetitions greater or equal 1*
- By configuration on sender- and on receiver-side, it is possible to add separately for each signal and/or separately for each signal group at most one update-bit.
- **ComUpdateBitPosition** -> position of the update-bit
- The AUTOSAR COM module shall handle the update-bit only internally and not as part of the signal or signal group
- The update-bits are not directly visible or accessible by AUTOSAR Software Compo-nents.

Update-BIT	
0	cleared/ data has not been updated
1	set/ data has been updated

Table 7 update-bit interpretation

Sender Side

Com user calls
Com_SendSignal -> update bit is set for signal
Com_SendSignalGroup -> update bit is set for signal group

ComTxIPduClearUpdateBit -> Transmit

All update-bits of all contained signals and signal groups are cleared after this I-PDU was sent out via PduR_ComTransmit(E_OK)

ComTxIPduClearUpdateBit -> Confirmation

All update-bits of all contained signals and signal groups are cleared after this I-PDU was sent out via PduR_ComTransmit(E_OK) and the I-PDU was successfully confirmed.

ComTxIPduClearUpdateBit -> TriggerTransmit

All update-bits of all contained signals and signal groups are cleared after the contents of this I-PDU was successfully requested by Com_TriggerTransmit.

Signal Gateway

- The signal gateway is an integrated part of the AUTOSAR COM module. Any external modules, except the cyclic task call, cannot access the signal gateway.
- The AUTOSAR COM module's signal gateway works with signals and signal groups.
- The AUTOSAR COM module's signal gateway supports only static routing for the **ComGwMapping** configuration container It is only possible to configure routes independently of the contents of the routed signals and group signals.
- The destination of a signal or group signal can be configured either per **ComGwDestinationDescription** container or via **ComGwSignal** reference
- The transmission mode will be selected based on the TMCs of other associated signals of the I-PDU. In case no signal contributes to the TMS calculation, the transmission mode **ComTxModeTrue** will be used
- The **ComGwMapping** configuration allows to configure routing a signal/ group signal from one source signal/ group signal to zero (no signal gateway functionality) or more destinations (1:n).

- **The Signal Gateway shall support the following I-PDU/ signal-processing stages on the receiver side**

1) reset reception deadline monitoring timer for I-PDU based monitoring

2) I-PDU callout

3) check of update-bits

4) endianness conversion and sign extension

- **The Signal Gateway shall support the following I-PDU/ signal-processing stages on the sender side**

1) set of update-bits

2) endianness conversion and sign extension

3) I-PDU transmission mode selection

4) I-PDU callout

Dealing with Signals

- The AUTOSAR COM module shall forward signals to be routed from received I-PDUs to transmit I-PDUs.
- **Configuration container -> ComGwMapping**
- Stopping the receive I-PDU after the signals to be routed have been forwarded to the transmit I-PDU will not prevent the transmit I-PDU from being transmitted unless it is stopped as well.

Dealing with Group Signals

- COM module shall forward routed group sig-nals from received I-PDUs to transmit I-PDUs COM module shall route all group signals be-longing to the same signal group in a consistent manner
- COM module shall support routing consistently a subset of group signals of a source signal group into a reduced target signal group.
- COM module shall support routing of individual group signals of a source signal group to a individual target signals (which are not part of signal group).

Routing of Timed Out Signals and Signal Groups

- COM module's signal gateway shall route signal and group signals even if any configured reception deadline monitoring timeout expired.
- In case of a not in time received signal or group signal the AUTOSAR COM module's signal gateway will route these signal or group signal anyway

Handling of Update-Bits

➢ If both, the received signal/ signal group and the destination signal/ signal group have an update-bit (ComUpdateBitPosition) :

- update-bit of the received signal/ signal group is set COM module shall route the signal/ signal group with the set update-bit and clear the update-bit of the destination signal/ signal group after it was sent.
- the update-bit of the received signal/ signal group is not set, the AUTOSAR COM module shall not route this signal/ signal group.

➢ If the received signal/ signal group has an update-bit but the destination signal has no update-bit and the update-bit is not set, the AUTOSAR COM module shall not route this signal/ signal group.

➢ If the received signal/ signal group has no update-bit and the destination signal/ signal group has an update-bit then COM module shall set the update-bit of the destination signal when a new signal/ signal group was received and clear it after sending of the destination signal/ signal group

Decoupling Signal Gateway

- To protect interrupt routines used for I-PDU reception from incalculable and perhaps expensive time usage, it is necessary to decouple the AUTOSAR COM module's signal gateway from interrupt routines.

- The AUTOSAR COM module shall execute all functions of its signal gateway during the function call Com_MainFunctionRouteSignals

- During this function call, the AUTOSAR COM module's signal gateway shall check received routed signals and group signals and copy them from the related receive I-PDUs to the related transmit I-PDUs

- Within Com_MainFunctionRouteSignals, the COM module shall evaluate the transfer properties and transmission modes and perform its routing actions in the following sequence:

  1. copy all gated signals from the source to the target I-PDUs
  2. evaluate the TMC of all gated signals
  3. evaluate the TMS for the target I-PDUs

Development Errors

Type of error	*Related error code*	*Error value*
API service called with wrong parameter	COM_E_ PARAM	0x01
Error code if any API service (except Com_GetStatus, Com_GetVersion Info, Com_MainFunctionRx, Com_MainFunctionTx and Com_Main FunctionRouteSignals) is called before the AUTOSAR COM module was initialized with Com_Init or after a call to Com_DeInit	COM_E_ UNINIT	0x02
NULL pointer checking	COM_E_ PARAM_ POINTER	0x03
Invalid configuration set selection	COM_E_ INIT_FAILED	0x04

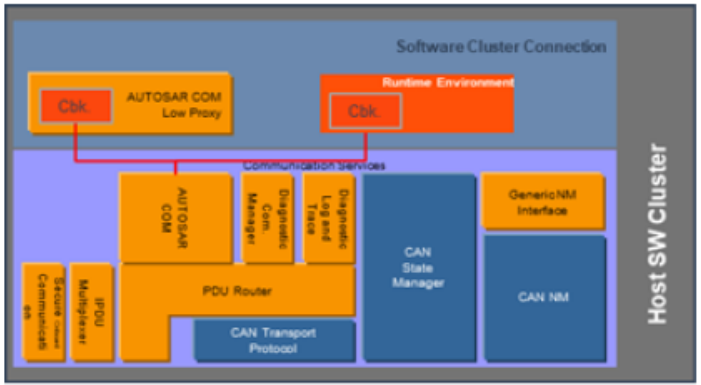
Runtime Errors

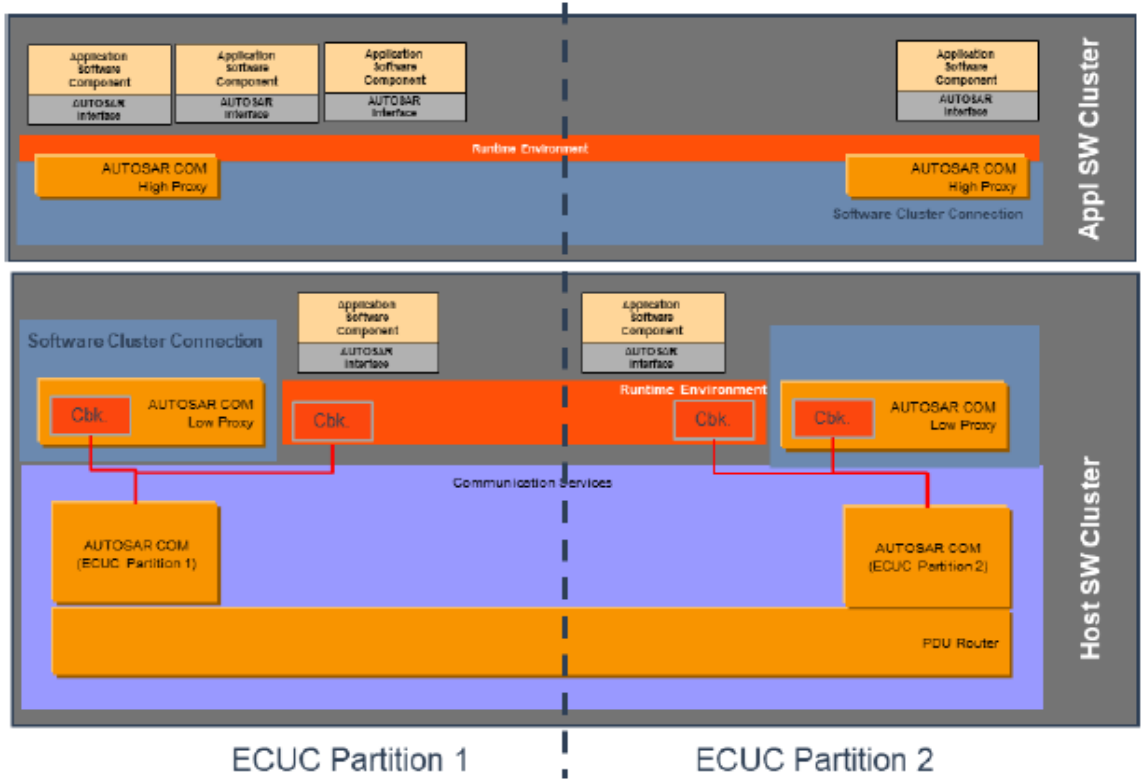
Type of error	*Related error code*	*Error value*
Transmission request was skipped	COM_E_SKIPPED_TRANSMISSION	0x05

Software Cluster Connection Layer (SwCluC)

- SwCluC groups the AUTOSAR embedded software into cluster (SWCL). It describes Host- and Applicative Cluster. Per machine, there is one Host-Cluster which contains most of or even all the BSW modules. The Applicative SWLC contain mostly application SW-C.

- From within a SWCL, the COM APIs are not accessed directly. Instead, accesses are done via a Proxy module (“Low Proxy”). For COM this means that it can have more than one upper layer users:
  - The RTE (of the Host SWCL)
  - The SwCluC Low Proxy
  - User defined




Application Software Component
AUTOSAR Interface
Application Software Component
AUTOSAR Interface
Application Software Component
AUTOSAR Interface
Application Software Component
AUTOSAR Interface
Runtime Environment
AUTOSAR COM High Proxy
AUTOSAR COM High Proxy
Software Cluster Connection
Appl SW Cluster
Software Cluster Connection
Application Software Component
AUTOSAR Interface
Application Software Component
AUTOSAR Interface
Runtime Environment
Cbk.
AUTOSAR COM Low Proxy
Cbk.
Cbk.
Cbk.
AUTOSAR COM Low Proxy
Communication Services
AUTOSAR COM (ECUC Partition 1)
AUTOSAR COM (ECUC Partition 2)
PDU Router
Host SW Cluster
ECUC Partition 1
ECUC Partition 2

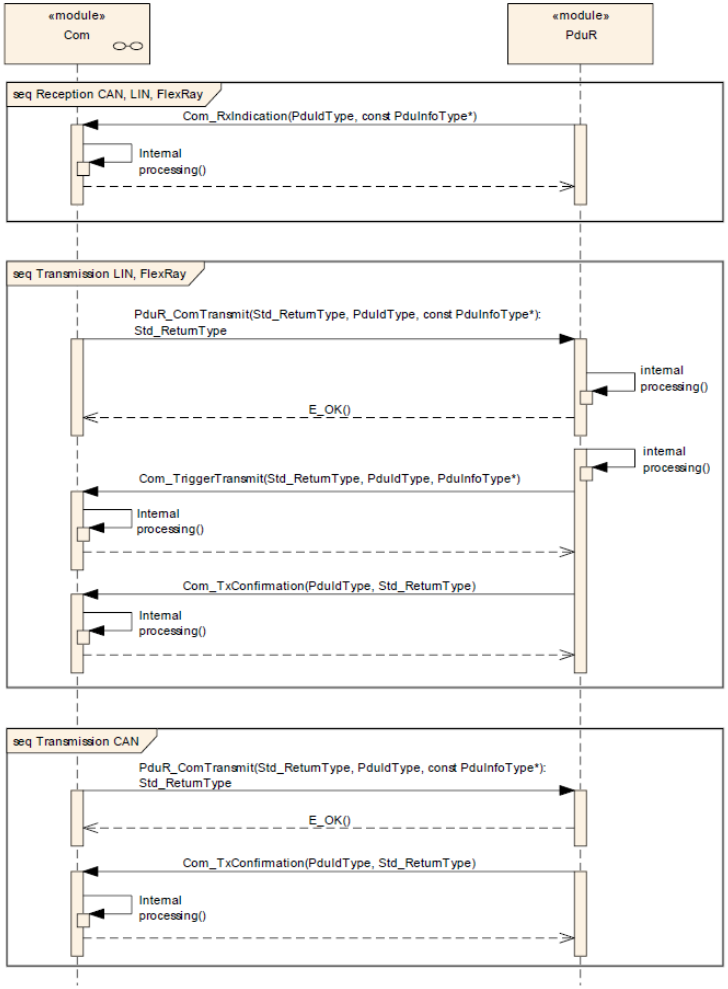
Receiver Side

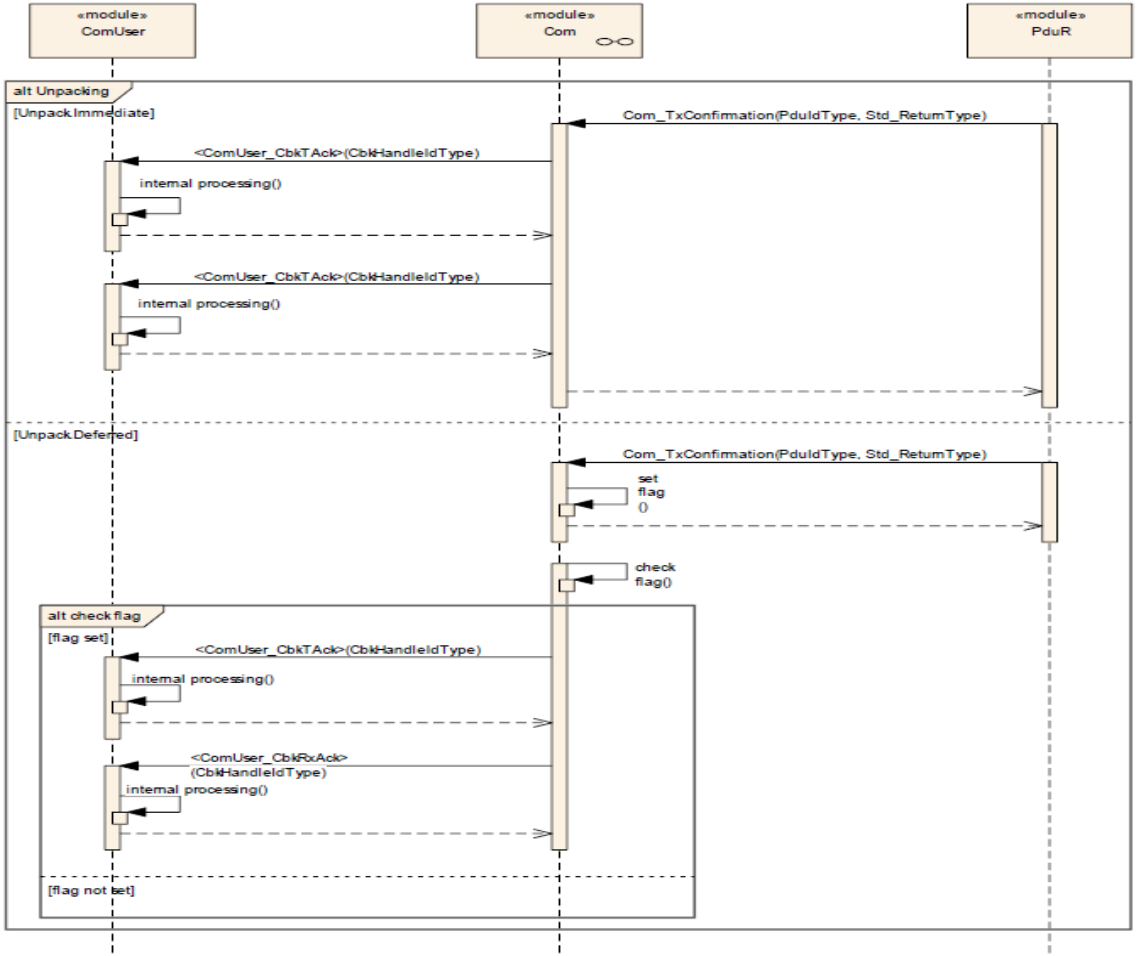
- **Update bit is set** -> If a signal or signal group with an attached update bit is received, the AUTOSAR COM module shall only process this signal, i.e. filter, notification, signal based gateway, byte swapping

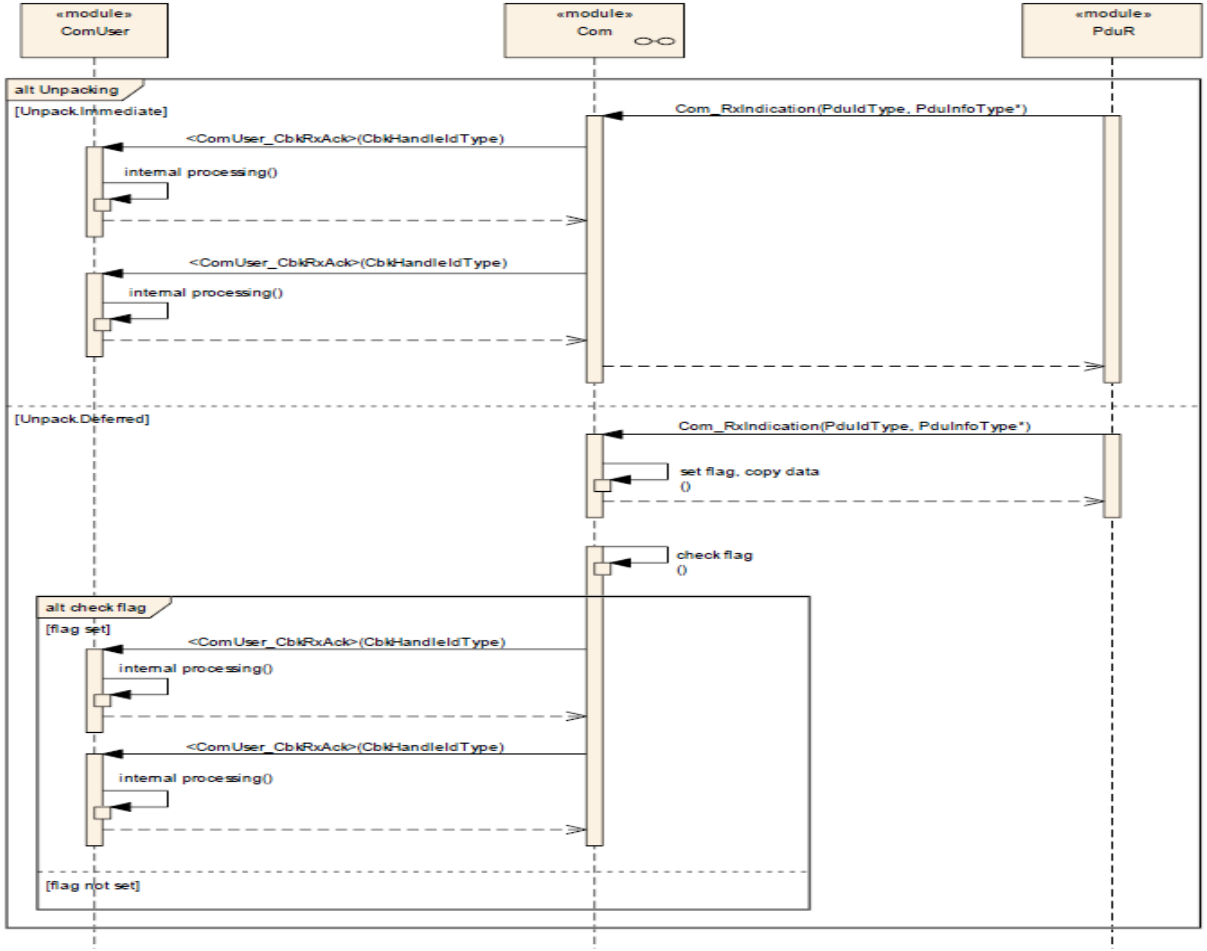
- **Update-bit is not set** -> If a signal or signal group with an attached update bit is received ,COM module shall discard this signal.

- If the signal has not been updated, the signal will not be routed via the signal gateway. It will only be discarded.

- The AUTOSAR COM module shall **interpret a signal/ signal group as updated** if **the signal has an update-bit** attached, and the value of the **update-bit is set**.


«module»
Com
«module»
PduR
seq Reception CAN, LIN, FlexRay
Com_RxIndication(PduIdType, const PduInfoType*)
Internal
processing()
seq Transmission LIN, FlexRay
PduR_ComTransmit(Std_ReturnType, PduIdType, const PduInfoType*):
Std_ReturnType
internal
processing()
E_OK()
internal
processing()
Com_TriggerTransmit(Std_ReturnType, PduIdType, PduInfoType*)
Internal
processing()
Com_TxConfirmation(PduIdType, Std_ReturnType)
Internal
processing()
seq Transmission CAN
PduR_ComTransmit(Std_ReturnType, PduIdType, const PduInfoType*):
Std_ReturnType
E_OK()
Com_TxConfirmation(PduIdType, Std_ReturnType)
Internal
processing()


«module»
ComUser
«module»
Com
«module»
PduR
alt Unpacking
[Unpack.Immediate]
Com_TxConfirmation(PduIdType, Std_ReturnType)
<ComUser_CbkTAck>(CbkHandleIdType)
internal processing()
<ComUser_CbkTAck>(CbkHandleIdType)
internal processing()
[Unpack.Deferred]
Com_TxConfirmation(PduIdType, Std_ReturnType)
set
flag
()
check
flag()
alt check flag
[flag set]
<ComUser_CbkTAck>(CbkHandleIdType)
internal processing()
<ComUser_CbkRxAck>
(CbkHandleIdType)
internal processing()
[flag not set]


«module»
ComUser
«module»
Com
«module»
PduR
alt Unpacking
[Unpack.Immediate]
Com_RxIndication(PduIdType, PduInfoType*)
<ComUser_CbkRxAck>(CbkHandleIdType)
internal processing()
<ComUser_CbkRxAck>(CbkHandleIdType)
internal processing()
[Unpack.Deferred]
Com_RxIndication(PduIdType, PduInfoType*)
set flag, copy data
()
check flag
()
alt check flag
[flag set]
<ComUser_CbkRxAck>(CbkHandleIdType)
internal processing()
<ComUser_CbkRxAck>(CbkHandleIdType)
internal processing()
[flag not set]

Function Definitions

Com_Init	void Com_Init (const Com_ConfigType* config)
Com_DeInit	void Com_DeInit (void)
Com_IpduGroupStart	void Com_IpduGroupStart (Com_IpduGroupIdType IpduGroupId, boolean initialize)
Com_IpduGroupStop	void Com_IpduGroupStop (Com_IpduGroupIdType IpduGroupId)
Com_EnableReceptionDM	void Com_EnableReceptionDM (Com_IpduGroupIdType IpduGroupId)
Com_DisableReceptionDM	void Com_DisableReceptionDM (Com_IpduGroupIdType IpduGroupId)
Com_GetStatus	Com_StatusType Com_GetStatus (void)
Com_GetVersionInfo	void Com_GetVersionInfo (Std_VersionInfoType* versioninfo)
Com_SendSignal	uint8 Com_SendSignal (Com_SignalIdType SignalId, const void* SignalDataPtr)
Com_SendDynSignal	uint8 Com_SendDynSignal (Com_SignalIdType SignalId, const void* SignalDataPtr, uint16 Length)

Com_ReceiveSignal	uint8 Com_ReceiveSignal (Com_SignalIdType SignalId, void* SignalDataPtr)
Com_ReceiveDynSignal	uint8 Com_ReceiveDynSignal (Com_SignalIdType SignalId, void* SignalDataPtr, uint16* Length)
Com_SendSignalGroup	uint8 Com_SendSignalGroup (Com_SignalGroupIdType SignalGroupId)
Com_ReceiveSignalGroup	uint8 Com_ReceiveSignalGroup (Com_SignalGroupIdType SignalGroupId)
Com_SendSignalGroupArray	uint8 Com_SendSignalGroupArray (Com_SignalGroupIdType SignalGroupId, const uint8* SignalGroupArrayPtr)
Com_ReceiveSignalGroupArray	uint8 Com_ReceiveSignalGroupArray (Com_SignalGroupIdType SignalGroupId, uint8* SignalGroupArrayPtr)
Com_ReceiveSignalWithMetaData	uint8 Com_ReceiveSignalWithMetaData (Com_SignalIdType SignalId, void* SignalDataPtr, uint8* MetaDataPtr)
Com_ReceiveDynSignalWithMetaData	uint8 Com_ReceiveDynSignalWithMetaData (Com_SignalIdType SignalId, void* SignalDataPtr, uint16* Length, uint8* MetaDataPtr)
Com_ReceiveSignalGroupWithMetaData	uint8 Com_ReceiveSignalGroupWithMetaData (Com_SignalGroupIdType SignalGroupId, uint8* MetaDataPtr)
Com_ReceiveSignalGroupArrayWithMetaData	void Com_ReceiveSignalGroupArrayWithMetaData (Com_SignalGroupIdType SignalGroupId, uint8* SignalGroupArrayPtr, uint8* MetaDataPtr)

Com_SendSignalWithMetaData	uint8 Com_SendSignalWithMetaData (Com_SignalIdType SignalId, const void* SignalDataPtr, const uint8* MetaDataPtr)
Com_SendDynSignalWithMetaData	uint8 Com_SendDynSignalWithMetaData (Com_SignalIdType SignalId, const void* SignalDataPtr, uint16 Length, const uint8* MetaDataPtr)
Com_SendSignalGroupWithMeta Data	uint8 Com_SendSignalGroupWithMetaData (Com_SignalGroupIdType SignalGroupId, const uint8* MetaDataPtr)
Com_SendSignalGroupArrayWith MetaData	uint8 Com_SendSignalGroupArrayWithMetaData (Com_SignalGroupIdType SignalGroupId, const uint8* SignalGroupArrayPtr, const uint8* MetaDataPtr)
Com_InvalidateSignal	uint8 Com_InvalidateSignal (Com_SignalIdType SignalId)
Com_InvalidateSignalGroup	uint8 Com_InvalidateSignalGroup (Com_SignalGroupIdType SignalGroupId)
Com_TriggerIPDUSend	Std_ReturnType Com_TriggerIPDUSend (PduIdType PduId)
Com_TriggerIPDUSendWithMeta Data	Std_ReturnType Com_TriggerIPDUSendWithMetaData (PduIdType PduId, const uint8* MetaData)
Com_SwitchIpduTxMode	void Com_SwitchIpduTxMode (PduIdType PduId, boolean Mode)

Callback Functions and Notifications

Com_TriggerTransmit	Std_ReturnType Com_TriggerTransmit (PduIdType TxPduId, PduInfoType* PduInfoPtr)
Com_RxIndication	void Com_RxIndication (PduIdType RxPduId, const PduInfoType* PduInfoPtr)
Com_TpRxIndication	void Com_TpRxIndication (PduIdType id, Std_ReturnType result)
Com_TxConfirmation	void Com_TxConfirmation (PduIdType TxPduId, Std_ReturnType result)
Com_TpTxConfirmation	void Com_TpTxConfirmation (PduIdType id, Std_ReturnType result)
Com_StartOfReception	BufReq_ReturnType Com_StartOfReception (PduIdType id, const PduInfoType* info, PduLengthType TpSduLength, PduLengthType* bufferSizePtr)
Com_CopyRxData	BufReq_ReturnType Com_CopyRxData (PduIdType id, const PduInfoType* info, PduLengthType* bufferSizePtr)
Com_CopyTxData	BufReq_ReturnType Com_CopyTxData (PduIdType id, const PduInfoType* info, const RetryInfoType* retry, PduLengthType* availableDataPtr)

Scheduled Functions

Com_MainFunctionRx	void Com_MainFunctionRx (void)
Com_MainFunctionTx	void Com_MainFunctionTx (void)
Com_MainFunctionRouteSignals	void Com_MainFunctionRouteSignals (void)

Mandatory Interfaces

API Function	Header File	Description
PduR_ComTransmit	PduR_Com.h	Requests transmission of a PDU.

Optional Interfaces

API Function	Header File	Description
Det_ReportError	Det.h	Service to report development errors.
Det_Report-RuntimeError	Det.h	Service to report runtime errors. If a callout has been configured then this callout shall be called.
PduR_ComCancel-Transmit	PduR_ Com.h	Requests cancellation of an ongoing transmission of a PDU in a lower layer communication module.

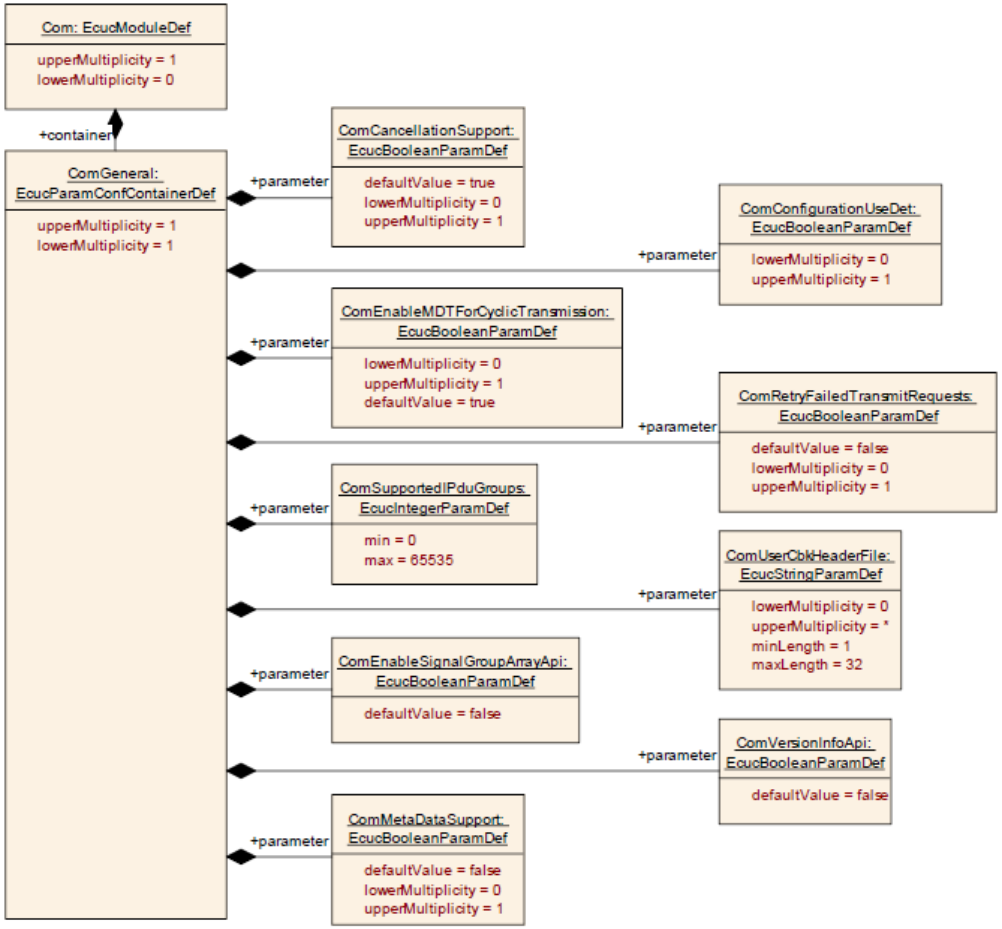
I-PDU Callout Functions

Com_RxIpduCallout	boolean Com_RxIpduCallout (PduIdType PduId, const PduInfoType* PduInfoPtr)
Com_TxIpduCallout	boolean Com_TxIpduCallout (PduIdType PduId, PduInfoType* PduInfoPtr)

Callback/ Notification Functions

<ComUser_CbkTxAck> (draft)	void <ComUser_CbkTxAck> (CbkHandleIdType ComUserCbkHandleId)
<ComUser_CbkTxErr> (draft)	void <ComUser_CbkTxErr> (CbkHandleIdType ComUserCbkHandleId)
<ComUser_CbkTxTOut> (draft)	void <ComUser_CbkTxTOut> (CbkHandleIdType ComUserCbkHandleId)
<ComUser_CbkRxAck> (draft)	void <ComUser_CbkRxAck> (CbkHandleIdType ComUserCbkHandleId)
<ComUser_CbkRxTOut> (draft)	void <ComUser_CbkRxTOut> (CbkHandleIdType ComUserCbkHandleId)
<ComUser_CbkInv> (draft)	void <ComUser_CbkInv> (CbkHandleIdType ComUserCbkHandleId)
Com_CbkTxPrep	void Com_CbkTxPrep (void)

ComGeneral


Com: EcucModuleDef
upperMultiplicity = 1
lowerMultiplicity = 0
+container
ComGeneral:
EcucParamConfContainerDef
upperMultiplicity = 1
lowerMultiplicity = 1
+parameter
ComCancellationSupport:
EcucBooleanParamDef
defaultValue = true
lowerMultiplicity = 0
upperMultiplicity = 1
+parameter
ComConfigurationUseDet:
EcucBooleanParamDef
lowerMultiplicity = 0
upperMultiplicity = 1
+parameter
ComEnableMDTForCyclicTransmission:
EcucBooleanParamDef
lowerMultiplicity = 0
upperMultiplicity = 1
defaultValue = true
+parameter
ComRetryFailedTransmitRequests:
EcucBooleanParamDef
defaultValue = false
lowerMultiplicity = 0
upperMultiplicity = 1
+parameter
ComSupportedIPduGroups:
EcucIntegerParamDef
min = 0
max = 65535
+parameter
ComUserCbkHeaderFile:
EcucStringParamDef
lowerMultiplicity = 0
upperMultiplicity = *
minLength = 1
maxLength = 32
+parameter
ComEnableSignalGroupArrayApi:
EcucBooleanParamDef
defaultValue = false
+parameter
ComVersionInfoApi:
EcucBooleanParamDef
defaultValue = false
+parameter
ComMetaDataSupport:
EcucBooleanParamDef
defaultValue = false
lowerMultiplicity = 0
upperMultiplicity = 1

ComGeneral

ComCancellationSupport	This parameter enables/disables the cancellation feature: true: enabled false: disabled
ComConfigurationUseDet	The error hook shall contain code to call the Det. If this parameter is configured COM_DEV_ERROR_DETECT shall be set to ON as output of the configuration tool.
ComEnableMDTForCyclicTransmission	Enables globally for the whole Com module the minimum delay time monitoring for cyclic and repeated transmissions
ComEnableSignalGroupArrayApi	Activate/Deactivate the signal group array access APIs (Com_SendSignalGroupArray, Com_ReceiveSignalGroupArray). true: signal group array access APIs activated false: signal group array access APIs deactivated
ComMetaDataSupport	This parameter enables/disables the support of meta-data feature including the API Com_TriggerIPDUSendWithMetaData. true: enabled false: disabled
ComRetryFailedTransmitRequests	If this Parameter is set to true, retry of failed transmission requests is enabled. If this Parameter is not present, the default value is assumed.
ComSupportedIPduGroups	Defines the maximum number of supported I-PDU groups.
ComUserCbkHeaderFile	Defines the header files for callback functions which shall be included by the COM module.
ComVersionInfoApi	Activate/Deactivate the version information API (Com_GetVersionInfo).

ComConfig

ComDataMemSize	Size of internal Com data in units of bytes (static memory allocation) - memory required by post-build configuration must be smaller than this constant.
ComMaxIPduCnt	Maximum number of IPdus. This parameter is needed only in case of post-build loadable implementation using static memory allocation.
ComGwMapping	Each instance of this container defines one mapping of the integrated Signal Gateway.
ComIPdu	if there is no ComIpdu container included no IPDUs are defined. In this case no communication via the AUTOSAR COM module is possible.
ComIPduGroup	if there is no ComIPduGroup container included then no IPDU group is defined
ComMainFunctionRouteSignals	Each element of this container defines one instance of Com_MainFunctionRouteSignals.
ComMainFunctionRx	Each element of this container defines one instance of Com_MainFunctionRx.
ComMainFunctionTx	Each element of this container defines one instance of Com_MainFunctionTx.
ComSignal	if there is no ComSignal container included no single signals are defined.
ComSignalGroup	if there is no ComSignalGroup container included no signal groups are defined
ComUserModule	Contains the configuration parameters of the Com user modules.

ComMainFunctionRx

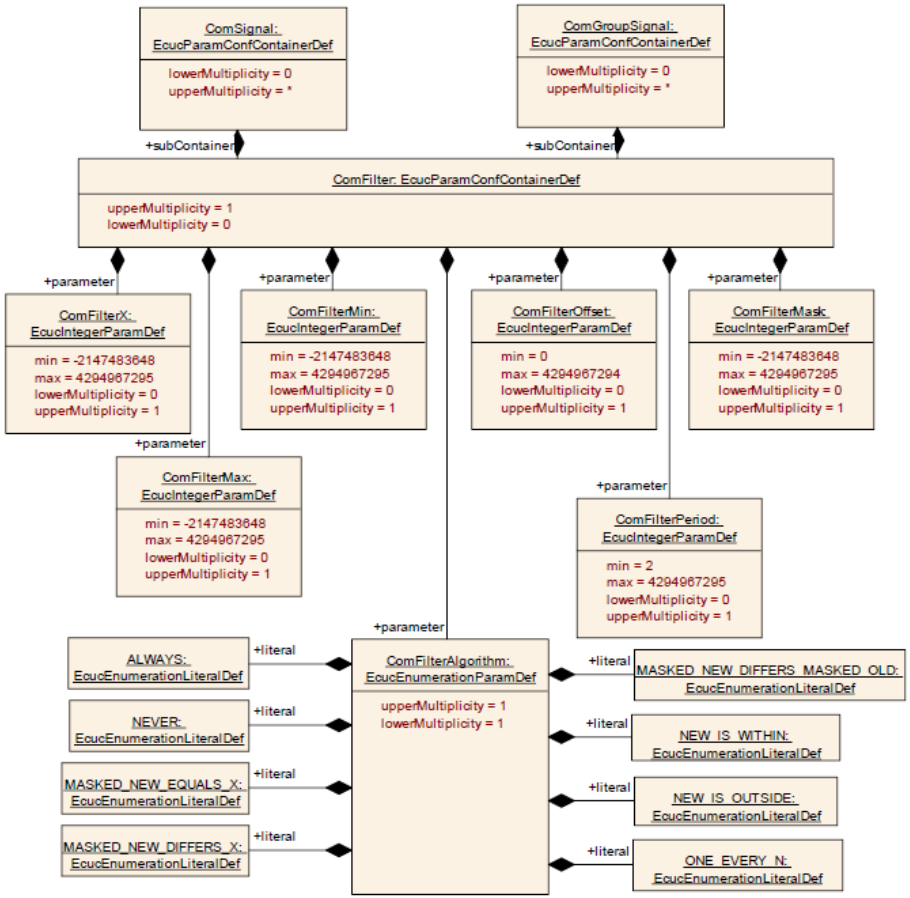
ComMainRxTimeBase	The period between successive calls to according instance of Com_MainFunctionRx in seconds. This parameter may be used by the COM generator to transform the values of the reception related timing configuration parameters of the COM module to internal implementation specific counter or tick values. The COM module's internal timing handling is implementation specific.
ComMainRxPartitionRef	Reference to EcucPartition, where the according Com_MainFunction instance is assigned to.

ComMainFunctionTx

ComMainTxTimeBase	The period between successive calls to according instance of Com_MainFunctionTx in seconds. This parameter may be used by the COM generator to transform the values of the reception related timing configuration parameters of the COM module to internal implementation specific counter or tick values. The COM module's internal timing handling is implementation specific.
ComPreparationNotification	This callback function indicates that the signals/signal groups to be sent via a dedicated Com_MainFunctionTx instance will now be prepared for transmission. If this parameter is omitted no notification shall take place.
ComMainTxPartitionRef	Reference to EcucPartition, where the according Com_MainFunction instance is assigned to.

ComMainFunctionRouteSignals

ComMainRouteSignalsTimeBase	The period between successive calls to according instance of Com_MainFunctionRouteSignals in seconds.
ComMainRouteSignalsPartitionRef	Reference to EcucPartition, where the according Com_MainFunction instance is assigned to.


ComSignal: EcucParamConfContainerDef
lowerMultiplicity = 0
upperMultiplicity = *
ComGroupSignal: EcucParamConfContainerDef
lowerMultiplicity = 0
upperMultiplicity = *
+subContainer
+subContainer
ComFilter: EcucParamConfContainerDef
upperMultiplicity = 1
lowerMultiplicity = 0
+parameter
ComFilterX: EcucIntegerParamDef
min = -2147483648
max = 4294967295
lowerMultiplicity = 0
upperMultiplicity = 1
+parameter
ComFilterMin: EcucIntegerParamDef
min = -2147483648
max = 4294967295
lowerMultiplicity = 0
upperMultiplicity = 1
+parameter
ComFilterOffset: EcucIntegerParamDef
min = 0
max = 4294967294
lowerMultiplicity = 0
upperMultiplicity = 1
+parameter
ComFilterMask: EcucIntegerParamDef
min = -2147483648
max = 4294967295
lowerMultiplicity = 0
upperMultiplicity = 1
+parameter
ComFilterMax: EcucIntegerParamDef
min = -2147483648
max = 4294967295
lowerMultiplicity = 0
upperMultiplicity = 1
+parameter
ComFilterPeriod: EcucIntegerParamDef
min = 2
max = 4294967295
lowerMultiplicity = 0
upperMultiplicity = 1
+parameter
ComFilterAlgorithm: EcucEnumerationParamDef
upperMultiplicity = 1
lowerMultiplicity = 1
+literal
ALWAYS: EcucEnumerationLiteralDef
+literal
NEVER: EcucEnumerationLiteralDef
+literal
MASKED_NEW_EQUALS_X: EcucEnumerationLiteralDef
+literal
MASKED_NEW_DIFFERS_X: EcucEnumerationLiteralDef
+literal
MASKED_NEW_DIFFERS_MASKED_OLD: EcucEnumerationLiteralDef
+literal
NEW_IS_WITHIN: EcucEnumerationLiteralDef
+literal
NEW_IS_OUTSIDE: EcucEnumerationLiteralDef
+literal
ONE_EVERY_N: EcucEnumerationLiteralDef

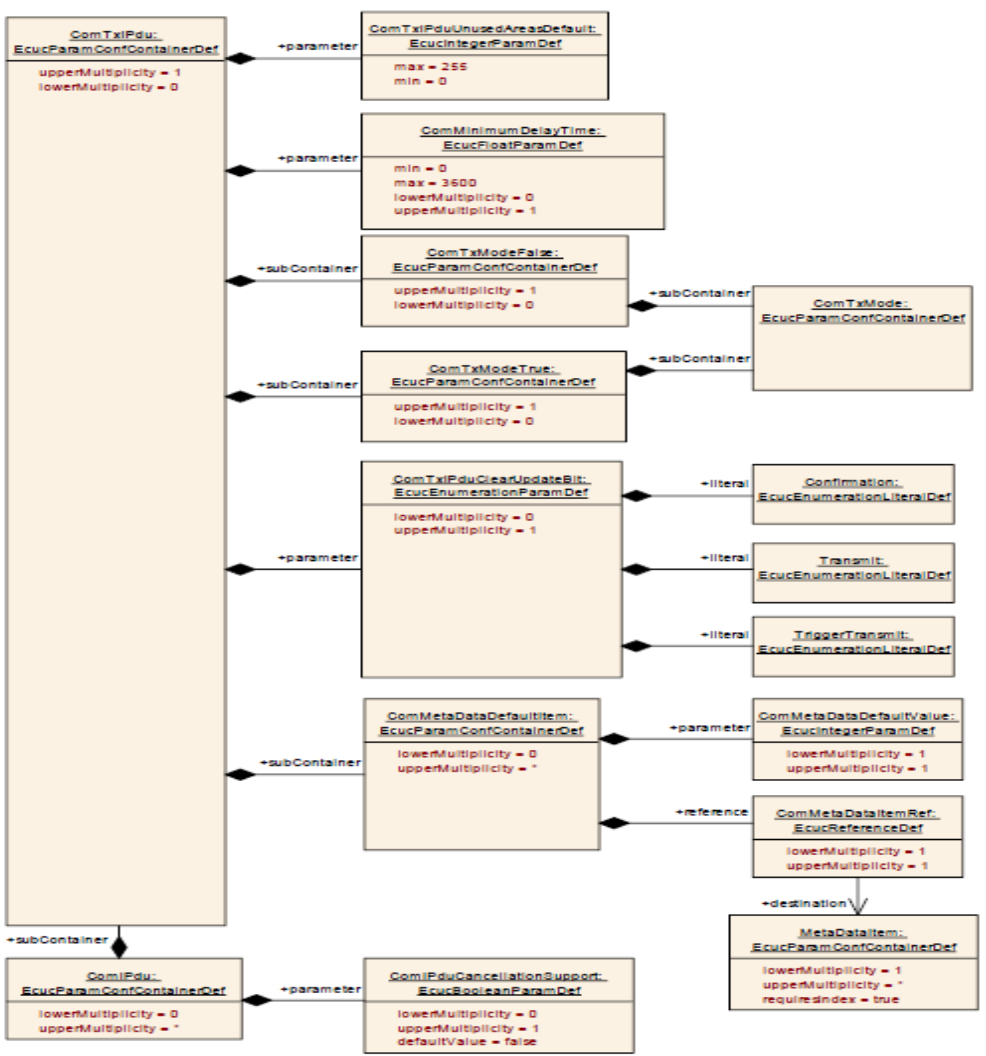
ComFilter

ComFilter	*Parent Container :* ComGroupSignal, ComGwDestinationDescription, ComSignal Note: On sender side the container is used to specify the transmission mode conditions.
ComFilterAlgorithm	ALWAYS, MASKED_NEW_DIFFERS_MASKED_OLD, MASKED_NEW_DIFFERS_X , MASKED_NEW_EQUALS_X ,NEVER, NEW_IS_OUTSIDE, NEW_IS_WITHIN, ONE_EVERY_N
ComFilterMask	The name of this attribute corresponds to the parameter name in the specification of Reception Filtering.
ComFilterMax	The name of this attribute corresponds to the max parameter name
ComFilterMin	The name of this attribute corresponds to the min parameter name
ComFilterOffset	The name of this attribute corresponds to the offset. Range = 0..(ComFilterPeriod-1)
ComFilterPeriod	This parameter defines the period of the ComFilterAlgorithm ONE_EVERY_N.

ComIPdu

ComIPduCallout	This parameter defines the existence and the name of a callout function for the corresponding I-PDU. If this parameter is omitted no I-PDU callout shall take place for the corresponding I-PDU.
ComIPduCancellationSupport	Defines for I-PDUs with ComIPduType NORMAL: If the underlying IF-modul supports cancellation of transmit requests. Defines for I-PDUs with ComIPduType TP: If the underlying TP-module supports RX and TX cancellation of ongoing requests.
ComIPduDirection	The direction defines if this I-PDU, and therefore the contributing signals and signal groups, shall be sent or received. Range: RECEIVE , SEND
ComIPduHandleId	The numerical value used as the ID of this I-PDU. The ComIPduHandleId is required by the API calls Com_RxIndication, Com_TpRxIndication, Com_StartOfReception and Com_CopyRxData to receive I-PDUs from the PduR . Etc.
ComIPduSignalProcessing	For the definition of the two modes Immediate and Deferred. Range: DEFERRED , IMMEDIATE
ComIPduTriggerTransmitCallout	If there is a trigger transmit callout defined for this I-PDU this parameter contains the name of the callout function.
ComIPduType	Defines if this I-PDU is a normal I-PDU that can be sent unfragmented or if this is a large I-PDU that shall be sent via the Transport Protocol of the underlying bus. Range : NORMAL , TP

ComIPduGroupRef	Reference to the I-PDU groups this I-PDU belongs to.
ComIPduMainFunctionRef	Reference to the Com_MainFunctionRx/Com_MainFunctionTx this I-PDU belongs to. Mandatory, if multiple main functions of the relevant type are defined.
ComIPduSignalGroupRef	References to all signal groups contained in this I-Pdu
ComIPduSignalRef	References to all signals contained in this I-PDU.
ComMainFunctionRouteSignalsRef	Reference to ComMainFunctionRouteSignals which performs signal gateway related activities.
ComPduIdRef	Reference to the "global" Pdu structure to allow harmonization of handle IDs in the COM-Stack.
ComTxIPdu	This container must be included if COM_IPDU_DIRECTION is configured to SEND.


ComTxIPdu:
EcucParamConfContainerDef
upperMultiplicity = 1
lowerMultiplicity = 0
+parameter
ComTxIPduUnusedAreasDefault:
EcucIntegerParamDef
max = 255
min = 0
+parameter
ComMinimumDelayTime:
EcucFloatParamDef
min = 0
max = 3600
lowerMultiplicity = 0
upperMultiplicity = 1
+subContainer
ComTxModeFalse:
EcucParamConfContainerDef
upperMultiplicity = 1
lowerMultiplicity = 0
+subContainer
ComTxMode:
EcucParamConfContainerDef
+subContainer
ComTxModeTrue:
EcucParamConfContainerDef
upperMultiplicity = 1
lowerMultiplicity = 0
+subContainer
+parameter
ComTxIPduClearUpdateBit:
EcucEnumerationParamDef
lowerMultiplicity = 0
upperMultiplicity = 1
+literal
Confirmation:
EcucEnumerationLiteralDef
+literal
Transmit:
EcucEnumerationLiteralDef
+literal
TriggerTransmit:
EcucEnumerationLiteralDef
+subContainer
ComMetaDataDefaultItem:
EcucParamConfContainerDef
lowerMultiplicity = 0
upperMultiplicity = *
+parameter
ComMetaDataDefaultValue:
EcucIntegerParamDef
lowerMultiplicity = 1
upperMultiplicity = 1
+reference
ComMetaDataItemRef:
EcucReferenceDef
lowerMultiplicity = 1
upperMultiplicity = 1
+destination
MetaDataItem:
EcucParamConfContainerDef
lowerMultiplicity = 1
upperMultiplicity = *
requiresIndex = true
+subContainer
ComIPdu:
EcucParamConfContainerDef
lowerMultiplicity = 0
upperMultiplicity = *
+parameter
ComIPduCancellationSupport:
EcucBooleanParamDef
lowerMultiplicity = 0
upperMultiplicity = 1
defaultValue = false

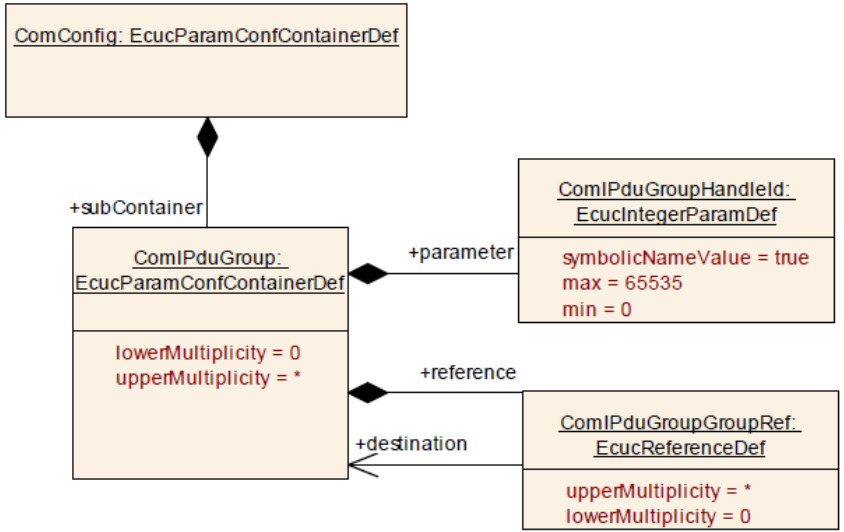
ComTxIPdu

ComMinimumDelayTime	Defines the Minimum Delay Time (MDT) between successive transmissions of this I-PDU in seconds.
ComTxIPduClearUpdateBit	Defines when the update-bits of signals or signal groups, contained in this I-PDU, will be cleared. Range: Confirmation , Transmit , TriggerTransmit
ComTxIPduUnusedAreasDefault	COM module fills not used areas of an I-PDU with this byte pattern. This attribute is mandatory to avoid undefined behaviour. This byte-pattern will be repeated throughout the I-PDU before any init-values or update-bits were set.
ComMetaDataDefaultItem	Defines a default value for a meta data item. Used for sending an I-PDU with meta data when it is triggered spontaneously (and not by Com_TriggerIPDUSendWithMetaData), and no meta data has been provided by the RTE. It represents a MetaDataItem of the referenced global PDU.
ComTxModeFalse	The referenced transmission mode object that is used when the filtering state for this I-PDU evaluates to false. The default is transmission mode None.
ComTxModeTrue	The referenced transmission mode object that is used when the filtering state for this I-PDU evaluates to true.

ComMetaDataDefaultItem

ComMetaDataDefaultValue	Default value for MetaDataItem of the global PDU.
ComMetaDataItemRef	Reference to a MetaDataItem of the global PDU.

ComIPduGroup



ComIPduGroupHandleId	The numerical value used as the ID of this I-PDU Group . The ComIPduGroupHandleId is required by the API calls to start and stop I-PDU Groups. Range: 0 .. (ComSupportedIPduGroups-1)
ComIPduGroupGroupRef	References to all I-PDU groups that includes this I-PDU group. If this reference is omitted this I-PDU group does not belong to another I-PDU group.

ComSignal

ComBitPosition	Starting position within the I-PDU. This parameter refers to the position in the I-PDU and not in the shadow buffer. If the endianness conversion is configured to Opaque the parameter ComBitPosition shall define the bit0 of the first byte like in little endian byte order
ComBitSize	Size in bits, for integer signal types. For ComSignalType UINT8_N and UINT8_DYN the size shall be configured by ComSignalLength. For ComSignalTypes FLOAT32 and FLOAT64 the size is already defined by the signal type and therefore may be omitted.
ComDataInvalidAction	This parameter defines the action performed upon reception of an invalid signal. Relating to signal groups the action in case if one of the included signals is an invalid signal. If Replace is used the ComSignalInitValue will be used for the replacement. Range : NOTIFY, REPLACE
ComFirstTimeout	Defines the length of the first deadline monitoring timeout period in seconds. This timeout is used immediately after start (or restart) of the deadline monitoring service.
ComHandleId	The numerical value used as the ID. This ID identifies signals and signal groups in the COM APIs using Com_SignalIdType or Com_SignalGroupIdType parameter respectively.
ComInitialValueOnly	This parameter defines that the respective signal's initial value shall be put into the respective PDU but there will not be any update of the value through the users (e.g. RTE, SwCluC).

ComInvalidNotification	Only valid on receiver side: Name of Com_CbkInv callback function to be called. Name of the function which notifies the RTE about the reception of an invalidated signal/ signal group. Only applicable if ComDataInvalidAction is configured to NOTIFY.
ComNotification	On sender side: Name of Com_CbkTxAck callback function to be called. On receiver side: Name of Com_CbkRxAck callback function to be called.
ComRxDataTimeoutAction	This parameter defines the action performed upon expiration of the reception deadline monitoring timer. Range : NONE, REPLACE, SUBSTITUTE
ComSignalDataInvalidValue	Defines the data invalid value of the signal.
ComSignalEndianness	Defines the endianness of the signal's network representation.
ComSignalInitValue	Initial value for this signal.
ComSignalLength	For ComSignalType UINT8_N this parameter specifies the length n in bytes. For ComSignalType UINT8_DYN it specifies the maximum length in bytes. For all other types this parameter shall be ignored.
ComSignalType	The AUTOSAR type of the signal. Whether or not the signal is signed or unsigned can be found by examining the value of this attribute. Range: BOOLEAN , FLOAT32 , FLOAT64, SINT16 , SINT32, SINT64, SINT8 , UINT16, UINT32, UINT64, UINT8, UINT8_DYN, UINT8_N

ComTimeout	Defines the length of the deadline monitoring timeout period in seconds.
ComTimeoutNotification	On sender side: Name of Com_CbkTxTOut callback function to be called. On receiver side: Name of Com_CbkRxTOut callback function to be called.
ComTimeoutSubstitutionValue	The signal substitution value will be used in case of a timeout and ComRxDataTimeoutAction is set to SUBSTITUTE.
ComTransferProperty	Defines if a write access to this signal can trigger the transmission of the corresponding I-PDU. If the I-PDU is triggered, depends also on the transmission mode of the corresponding I-PDU. Range : PENDING, TRIGGERED, TRIGGERED_ON_CHANGE , TRIGGERED_ON_CHANGE_WITHOUT_REPETITION, TRIGGERED_WITHOUT_REPETITION
ComUpdateBitPosition	Bit position of update-bit inside I-PDU.
ComSystemTemplateSystemSignalRef	Reference to the ISignalToIPduMapping that contains a reference to the ISignal (System Template) which this ComSignal (or ComGroupSignal) represents.

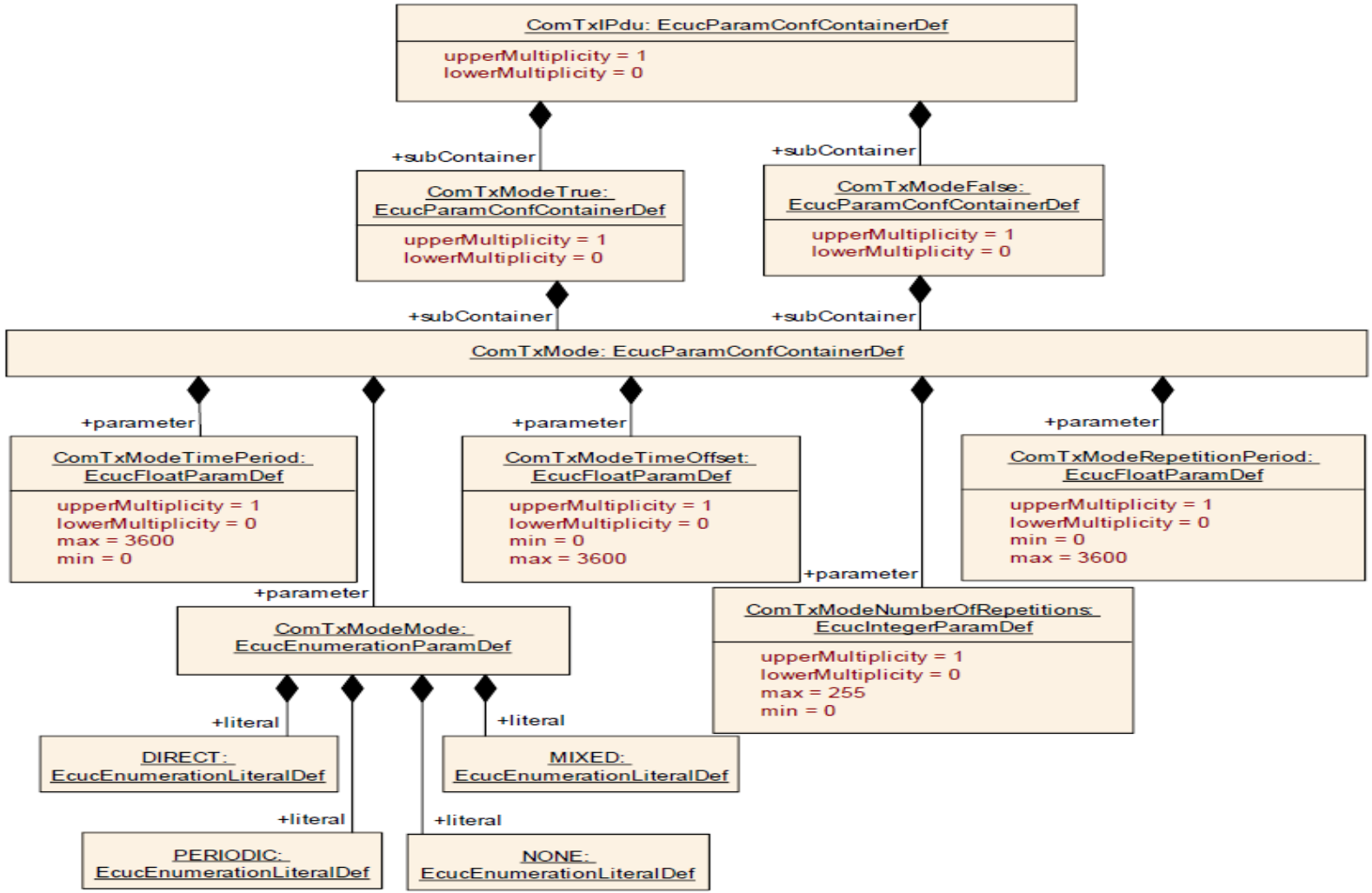
ComSignalGroup

ComDataInvalidAction	This parameter defines the action performed upon reception of an invalid signal. Range : NOTIFY , REPLACE
ComErrorNotification	Only valid on sender side: Name of Com_CbkTxErr callback function to be called.
ComFirstTimeout	Defines the length of the first deadline monitoring timeout period in seconds.
ComHandleId	The numerical value used as the ID. Used in Com_SignalIdType and Com_SignalGroupIdType
ComInitialValueOnly	This parameter defines that the respective signal's initial value shall be put into the respective PDU
ComInvalidNotification	Only valid on receiver side: Name of Com_CbkInv callback function to be called.
ComNotification	On sender side: Name of Com_CbkTxAck callback function to be called. On receiver side: Name of Com_CbkRxAck callback function to be called.
ComRxDataTimeoutAction	Defines the action performed upon expiration of the reception deadline monitoring timer. Range: NONE, REPLACE, SUBSTITUTE
ComSignalGroupArrayAccess	Defines whether the uint8-array based access shall be used for this ComSignalGroup.

ComTimeout	Defines the length of the deadline monitoring timeout period in seconds.
ComTimeoutNotification	On sender side: Name of Com_CbkTxTOut callback function to be called. On receiver side: Name of Com_CbkRxTOut callback function to be called.
ComTransferProperty	
ComUpdateBitPosition	Bit position of update-bit inside I-PDU.
ComSystemTemplateSignalGroupRef	Reference to the ISignalToIPduMapping that contains a reference to the ISignalGroup (SystemTemplate) which this ComSignalGroup represents.

ComGroupSignal

ComBitPosition	Starting position within the I-PDU.
ComBitSize	Size in bits, for integer signal types. For ComSignalType UINT8_N and UINT8_DYN
ComHandleId	The numerical value used as the ID.
ComSignalDataInvalidValue	Defines the data invalid value of the signal.
ComSignalEndianness	Defines the endianness of the signal's network representation.
ComSignalInitValue	Initial value for this signal.
ComSignalLength	For ComSignalType UINT8_N this parameter specifies the length n in bytes. For ComSignalType UINT8_DYN it specifies the maximum length in bytes.
ComSignalType	The AUTOSAR type of the signal. Whether or not the signal is signed or unsigned can be found by examining the value of this attribute.
ComTimeoutSubstitutionValue	The signal substitution value will be used in case of a timeout and ComRxDataTimeoutAction is set to SUBSTITUTE.
ComTransferProperty	PENDING / TRIGGERED_ON_CHANGE
ComSystemTemplateSystemSignalRef	Reference to the ISignalToIPduMapping that contains a reference to the ISignal (System Template) which this ComSignal (or ComGroupSignal) represents.


ComTxIPdu: EcucParamConfContainerDef
upperMultiplicity = 1
lowerMultiplicity = 0
+subContainer
+subContainer
ComTxModeTrue: EcucParamConfContainerDef
upperMultiplicity = 1
lowerMultiplicity = 0
ComTxModeFalse: EcucParamConfContainerDef
upperMultiplicity = 1
lowerMultiplicity = 0
+subContainer
+subContainer
ComTxMode: EcucParamConfContainerDef
+parameter
+parameter
+parameter
+parameter
ComTxModeTimePeriod: EcucFloatParamDef
upperMultiplicity = 1
lowerMultiplicity = 0
max = 3600
min = 0
ComTxModeTimeOffset: EcucFloatParamDef
upperMultiplicity = 1
lowerMultiplicity = 0
min = 0
max = 3600
ComTxModeRepetitionPeriod: EcucFloatParamDef
upperMultiplicity = 1
lowerMultiplicity = 0
min = 0
max = 3600
+parameter
+parameter
ComTxModeMode: EcucEnumerationParamDef
ComTxModeNumberOfRepetitions: EcucIntegerParamDef
upperMultiplicity = 1
lowerMultiplicity = 0
max = 255
min = 0
+literal
+literal
+literal
+literal
DIRECT: EcucEnumerationLiteralDef
MIXED: EcucEnumerationLiteralDef
PERIODIC: EcucEnumerationLiteralDef
NONE: EcucEnumerationLiteralDef

ComTxMode

ComTxModeMode	The available transmission modes shall be extended by the additional mode None. Range : DIRECT, MIXED, NONE, PERIODIC
ComTxModeNumberOfRepetitions	Defines the number of repetitions for the transmission mode DIRECT and the event driven part of transmission mode MIXED.
ComTxModeRepetitionPeriod	Defines the repetition period in seconds of the multiple transmissions in case ComTxModeNumberOfRepetitions is configured greater than or equal to 1 and ComTxModeMode is configured to DIRECT or MIXED.
ComTxModeTimeOffset	Defines the period in seconds between the start of the I-PDU by Com_IpduGroupStart and the first transmission request in case ComTxModeMode is configured to PERIODIC or MIXED.
ComTxModeTimePeriod	Defines the repetition period in seconds of the periodic transmission

ComTxModeTrue	This container contains the configuration parameters of the AUTOSAR COM module's transmission modes in the case the ComFilter evaluates to true.
ComTxModeFalse	This container contains the configuration parameters of the AUTOSAR COM module's transmission modes in the case the ComFilter evaluates to false.

ComGwMapping

ComGwMapping	Each instance of this container defines one mapping of the integrated Signal Gateway.
ComGwDestination	Each instance of this choice container allows to define one routing destination either by reference to an already configured COM signal / group signal or by a destination description container.
ComGwSource	This choice container allows the definition of the gateway source signal either by reference to an already configured COM signal / group signal or by a source description container.

ComGwSource

ComGwSignal	This container allows specifying a gateway source or destination respectively with a reference to a ComSignal or a ComGroupSignal.
ComGwSourceDescription	Description of a gateway source. This container allows defining a gateway source without the configuration of a complete COM signal. This allows adding / changing gateway relations post build without the configuration of new signals.

ComGwSourceDescription

ComBitPosition	Starting position within the I-PDU.
ComBitSize	Size in bits, for integer signal types.
ComSignalEndianness	Defines the endianness of the signal's network representation.
ComSignalLength	specifies the length n in bytes.
ComSignalType	The AUTOSAR type of the signal.
ComUpdateBitPosition	Bit position of update-bit inside I-PDU.
ComGwIPduRef	Reference to an I-PDU of a Signal Gateway source or destination description.

ComGwDestination

ComGwDestinationDescription	Description of a gateway destination. This container allows defining a gateway destination without the configuration of a complete COM signal. This allows adding / changing gateway relations post build without the configuration of new signals.
ComGwSignal	This container allows specifying a gateway source or destination respectively with a reference to a ComSignal or a ComGroupSignal.

ComUserModule

ComUserModule	Contains the configuration parameters of the Com user modules.
ComUserModuleCnfRef	Reference to the Com user module configuration.

ComUserModuleCnf

ComUserHeaderInclude	Defines the header file where the Com user provides the function declarations for configured callbacks.
ComUserCallback	This container defines a Com callback function for signals and signal groups.
ComUserSignal	Contains the configuration parameters of a signal inside a Com user module.
ComUserSignalGroup	Contains the configuration parameters of a signal group inside a Com user module.


thank
you